

1.a) Program to Link JavaScript file with the HTML page

Index.html

```
<!DOCTYPE html>

<html>

<head>

    <title>JavaScript Linking Example</title>

</head>

<body>

    <h1>JavaScript Link Test</h1>

    <button onclick="showMessage()">Click Me</button>

    <script src="script.js"></script>

</body>

</html>
```

Script.js

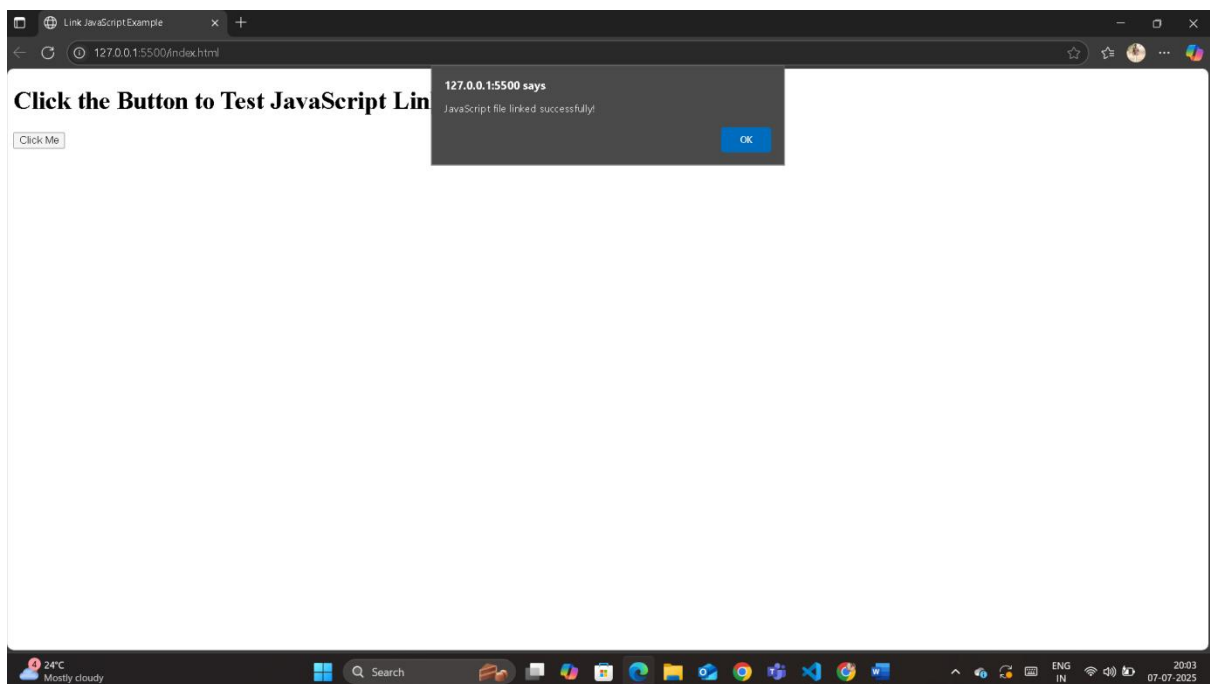
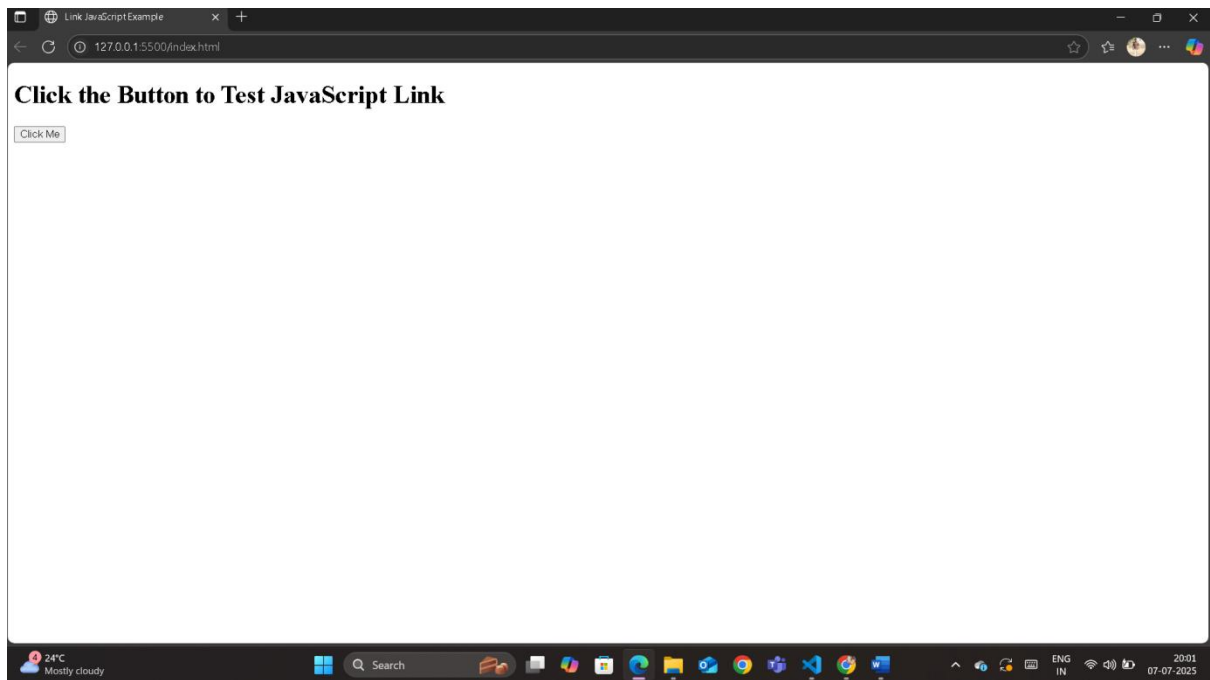
```
// This is an external JavaScript file

function showMessage() {

    alert("JavaScript file linked successfully!");

}
```

Output:



1.b)Program to Select the elements in HTML page using Selectors

HTML CODE

```
<!DOCTYPE html>

<html>

<head>

  <title>DOM Selectors Example</title>

</head>

<body>

  <h1 id="main-heading">Welcome to DOM Selectors</h1>

  <p class="description">This is a paragraph with class 'description'.</p>

  <p class="description">Another paragraph with same class.</p>

  <button id="clickMe">Click Me</button>

  <!-- Link to external JavaScript file -->

  <script src="script.js"></script>

</body>

</html>
```

JAVASCRIPT CODE

```
const heading = document.getElementById("main-heading");
heading.style.color = "blue";

const descriptions = document.getElementsByClassName("description");
for (let i = 0; i < descriptions.length; i++) {
  descriptions[i].style.fontWeight = "bold";
}

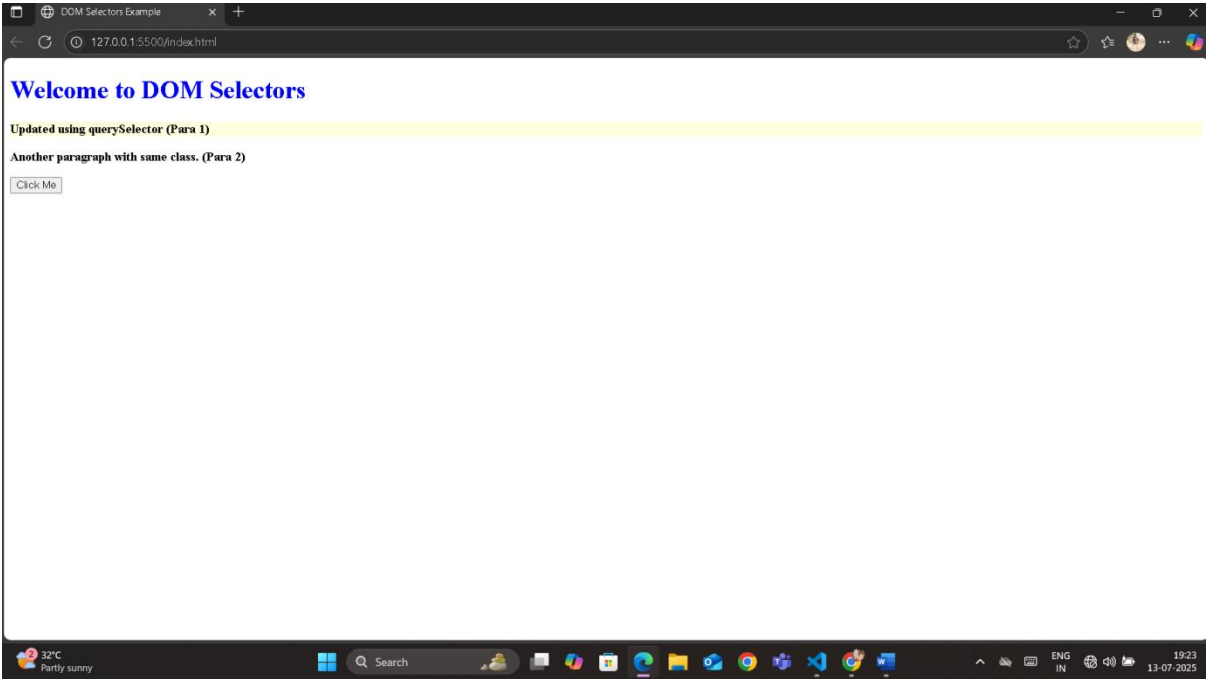
const paragraphs = document.getElementsByTagName("p");
paragraphs[0].style.backgroundColor = "lightyellow";
```

```
const firstPara = document.querySelector(".description");
firstPara.innerText = "Updated using querySelector";

const allParas = document.querySelectorAll(".description");
allParas.forEach((p, index) => {
  p.innerText += ` (Para ${index + 1})`;
});

console.log("Heading Element:", heading);
console.log("Descriptions:", descriptions);
console.log("Paragraphs:", paragraphs);
console.log("First .description:", firstPara);
console.log("All .description:", allParas);
```

Output:



1.c) Program to Implement the events listeners

HTML CODE

```
<!DOCTYPE html>

<html>

<head>

  <title>Event Listeners Example</title>

</head>

<body>

  <h1>JavaScript Event Listeners Demo</h1>

  <button id="myButton">Click Me</button>

  <p id="outputText">This is some text.</p>

  <script src="script.js"></script>

</body>

</html>
```

JAVASCRIPT CODE

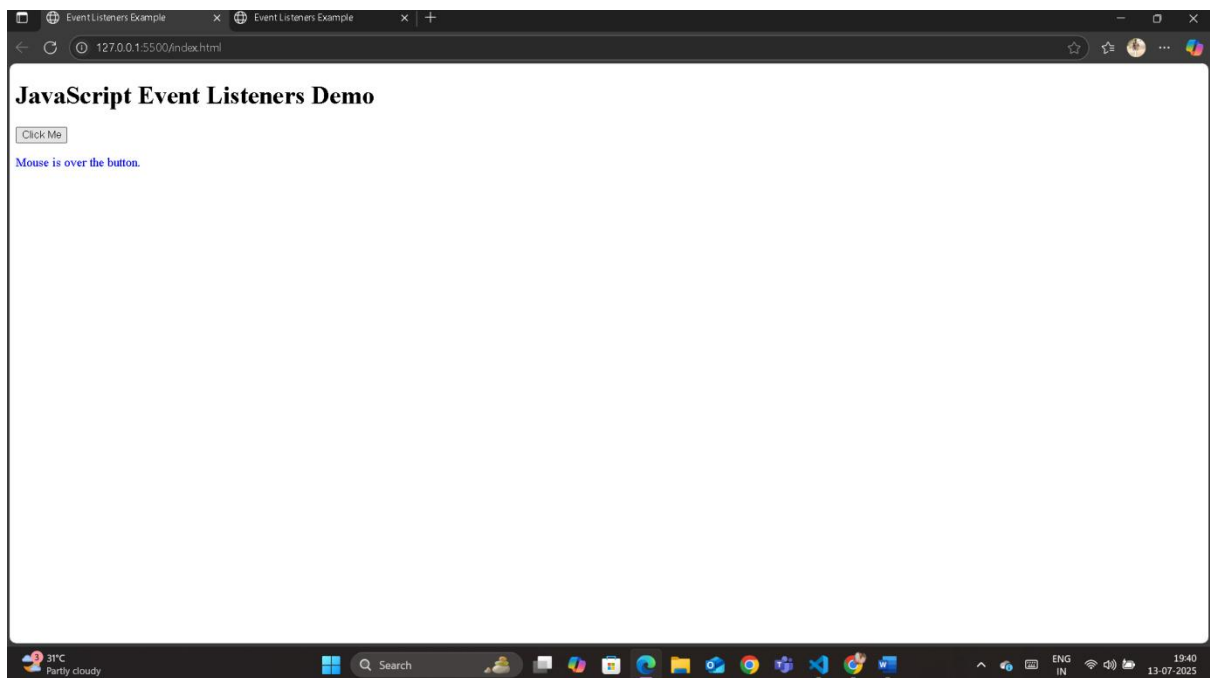
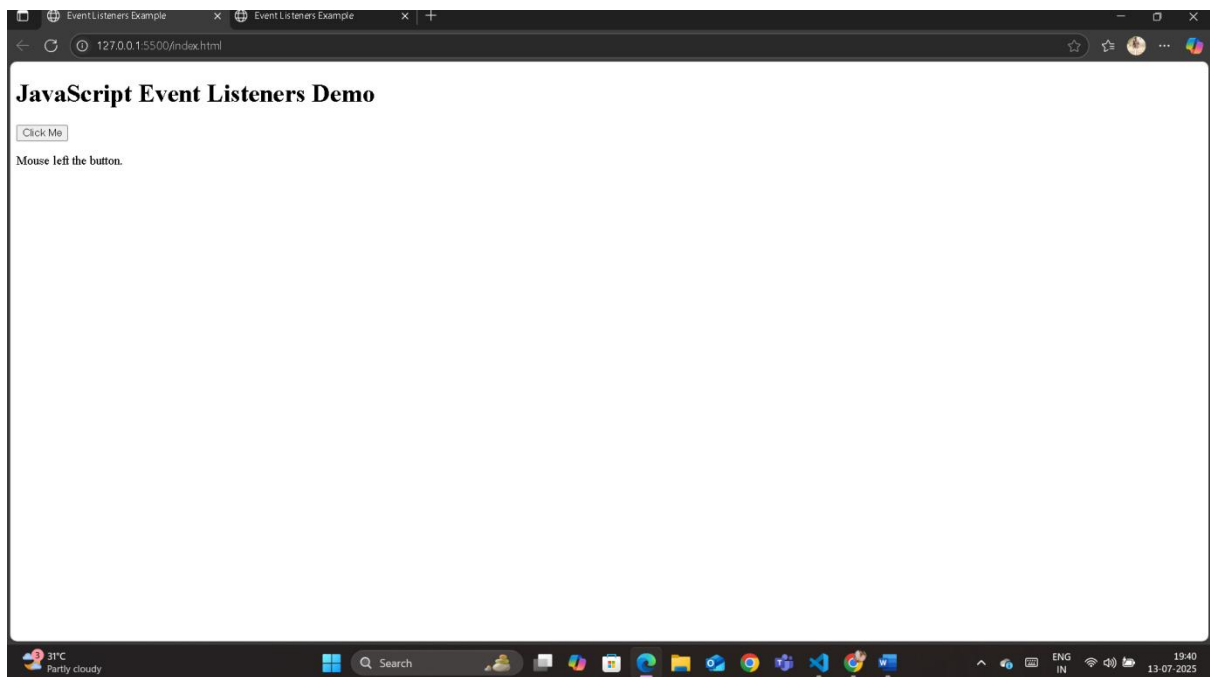
```
const button = document.getElementById("myButton");
const output = document.getElementById("outputText");

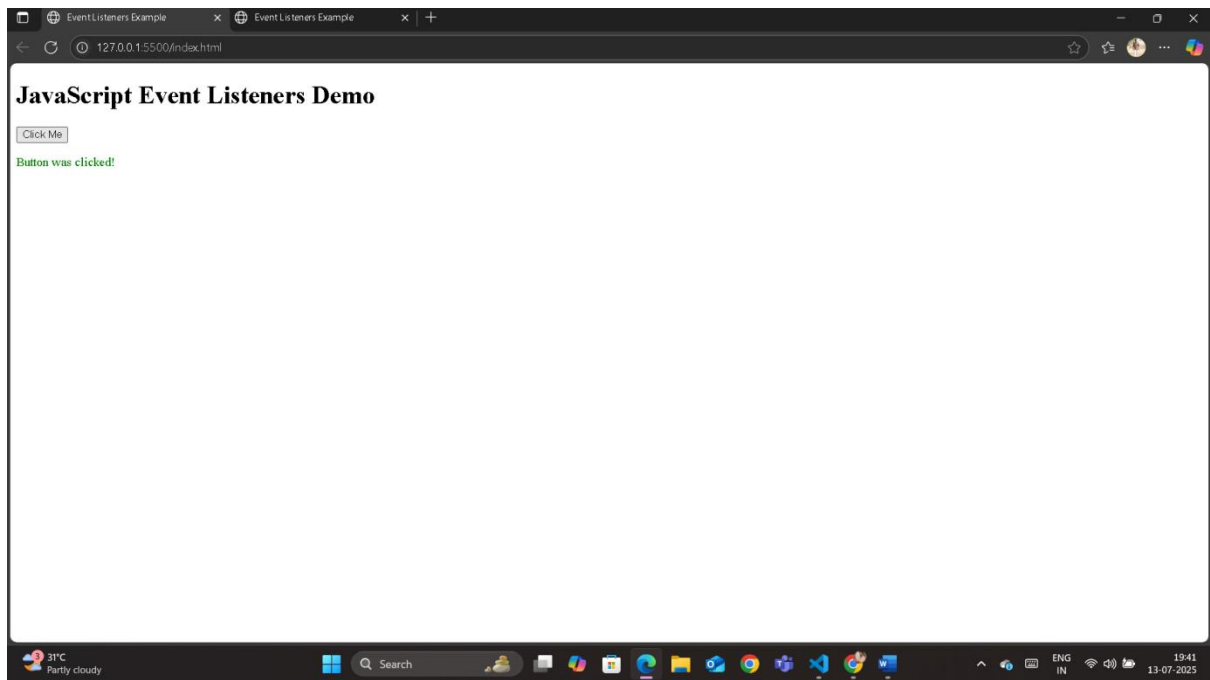
button.addEventListener("click", function () {
  output.innerText = "Button was clicked!";
  output.style.color = "green";
});

button.addEventListener("mouseover", function () {
  output.innerText = "Mouse is over the button.";
  output.style.color = "blue";
});
```

```
button.addEventListener("mouseout", function () {  
    output.innerText = "Mouse left the button."  
    output.style.color = "black";  
});
```

Output:





1.d) Program to Handle the click events for the HTML button elements

```
<!DOCTYPE html>

<html>

<head>

  <title>Button Click Event</title>

  <style>

    button {

      margin: 10px;

      padding: 10px 20px;

      font-size: 16px;

    }

    #output {

      margin-top: 20px;

      font-size: 18px;

      color: green;

    }

  </style>

</head>

<body>

  <h2>Click Event Handler Example</h2>

  <button id="btn1">Click Me 1</button>

  <button id="btn2">Click Me 2</button>

  <button id="btn3">Click Me 3</button>

  <div id="output"></div>

  <!-- Link to external JavaScript file -->

  <script src="script.js"></script>

</body>

</html>
```

```
//Script.JS Code
```

```
// Function to display a message in the output div
```

```
function handleClick(message) {  
    document.getElementById("output").innerText = message;  
}
```

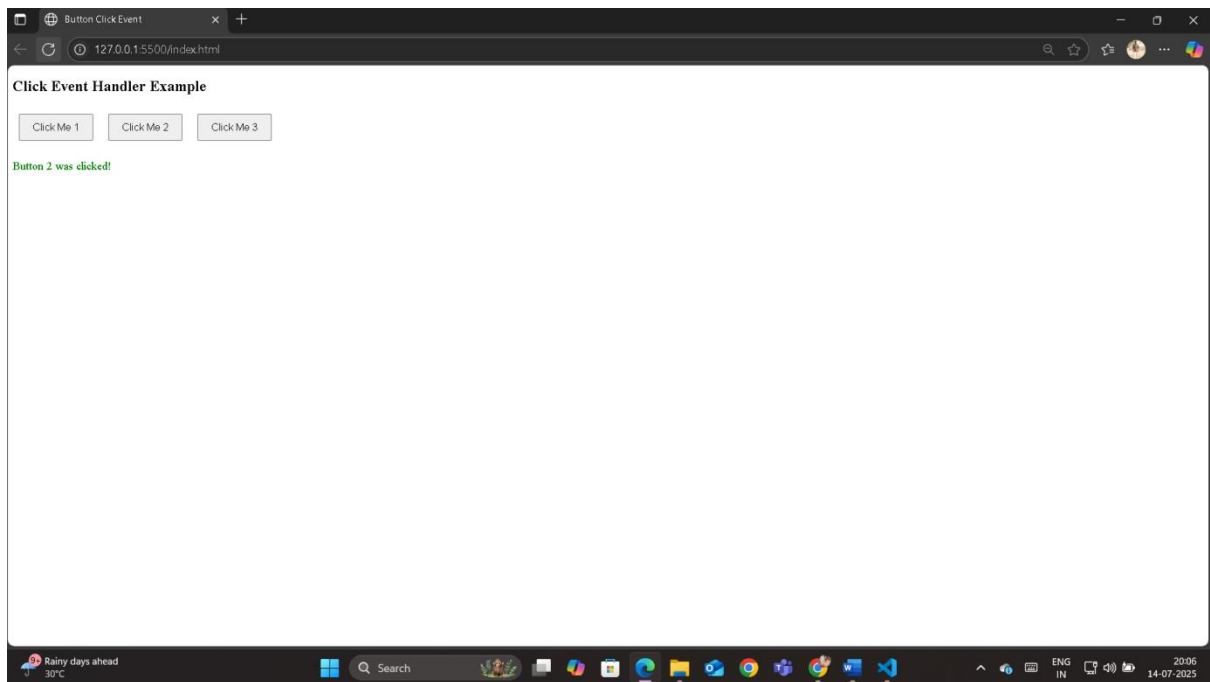
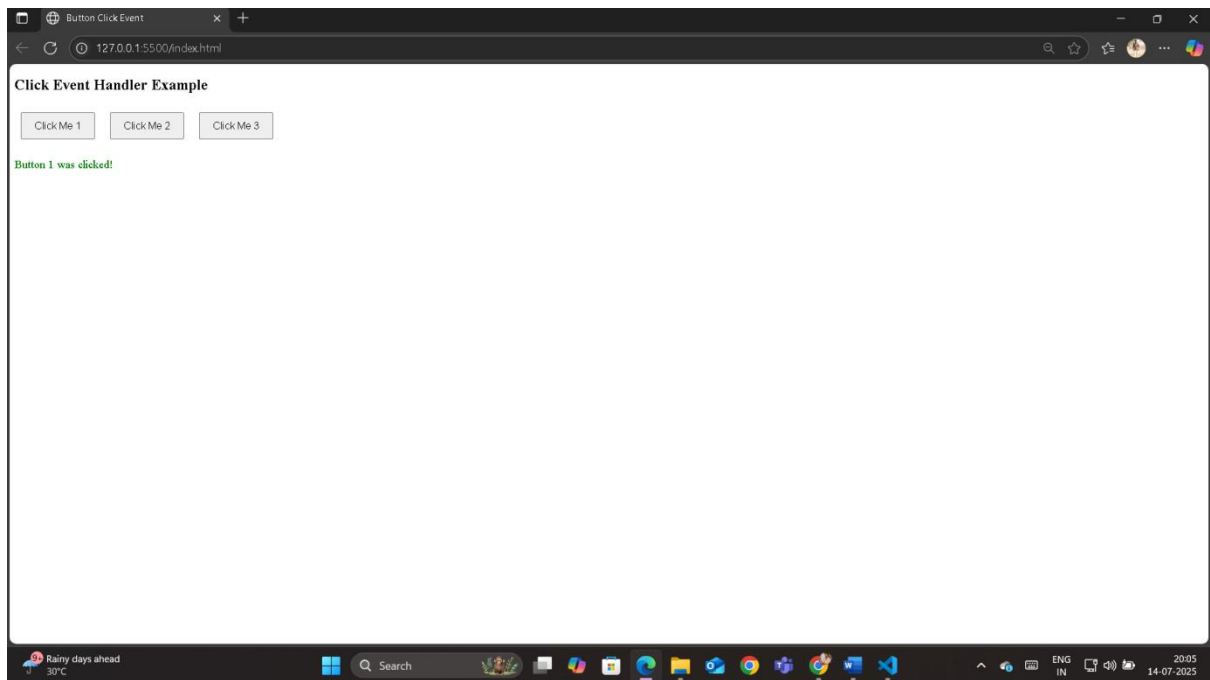
```
// Attach event listeners to buttons
```

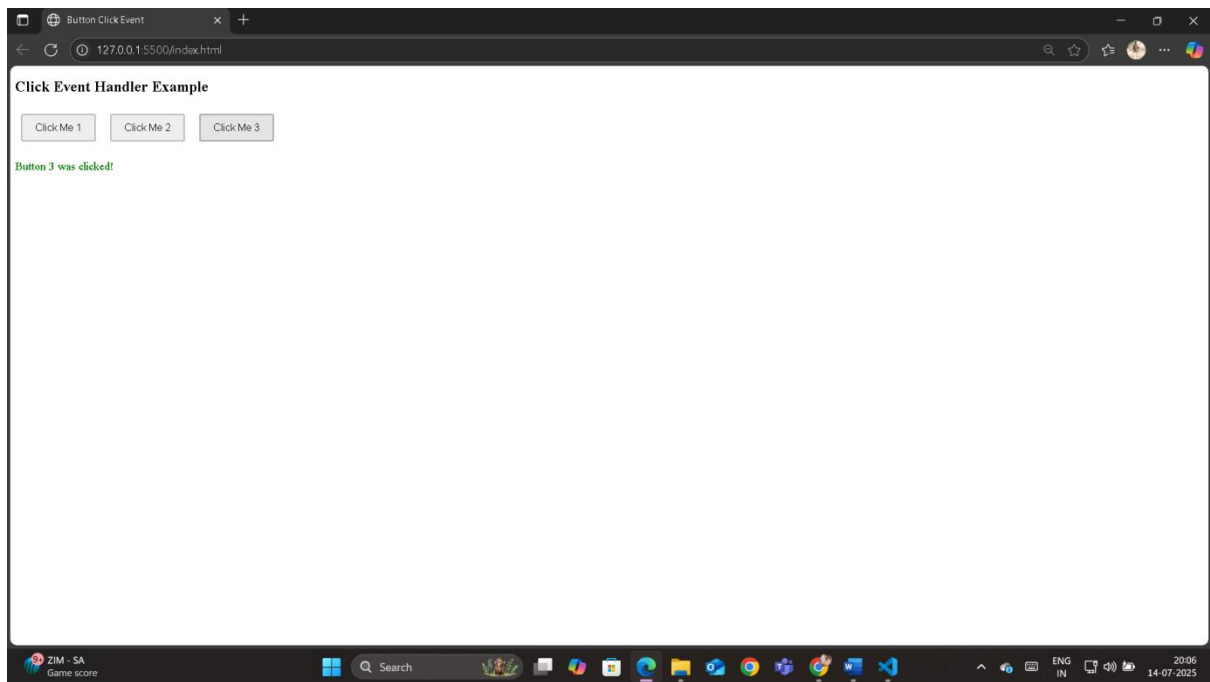
```
document.getElementById("btn1").addEventListener("click", function () {  
    handleClick("Button 1 was clicked!");  
});
```

```
document.getElementById("btn2").addEventListener("click", function () {  
    handleClick("Button 2 was clicked!");  
});
```

```
document.getElementById("btn3").addEventListener("click", function () {  
    handleClick("Button 3 was clicked!");  
});
```

OUTPUT:





1. e) Program to With three types of Functions

// 1. Function Declaration

```
function add(a, b) {  
    return a + b;  
}
```

// 2. Function Definition using Function Expression

```
const subtract = function(a, b) {  
    return a - b;  
};
```

// 3. Arrow Function

```
const multiply = (a, b) => a * b;
```

// Function Calls

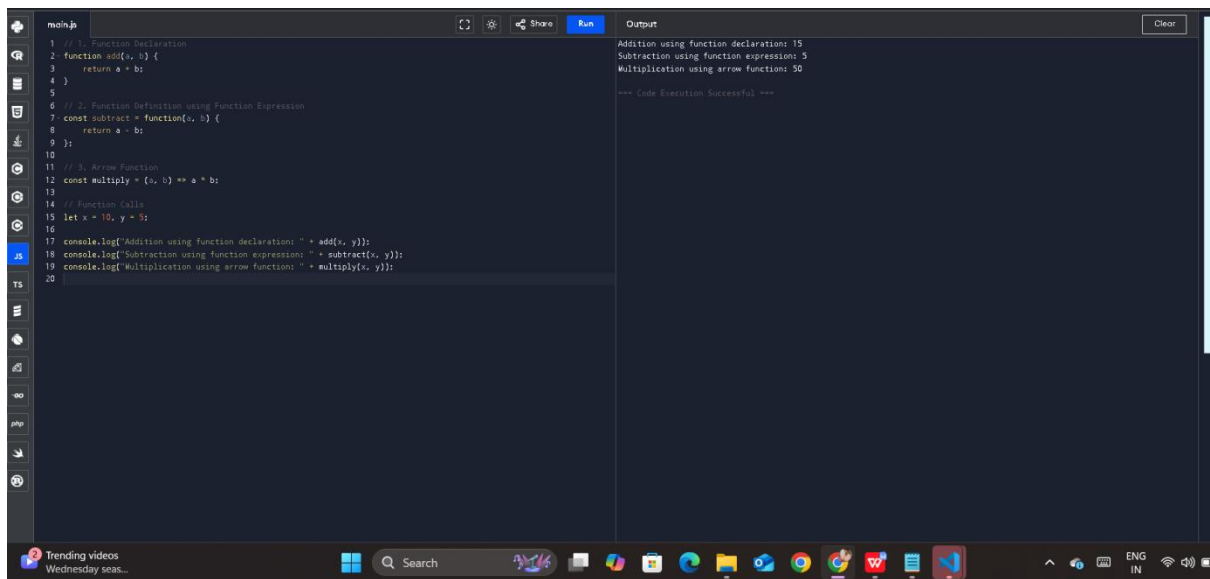
```
let x = 10, y = 5;
```

```
console.log("Addition using function declaration: " + add(x, y));
```

```
console.log("Subtraction using function expression: " + subtract(x, y));
```

```
console.log("Multiplication using arrow function: " + multiply(x, y));
```

Output:



The screenshot shows a web-based JavaScript IDE with a dark theme. The editor on the left contains the following code:

```
1 // 1. Function Declaration
2 function add(a, b) {
3   return a + b;
4 }
5
6 // 2. Function Definition using Function Expression
7 const subtract = function(a, b) {
8   return a - b;
9 };
10
11 // 3. Arrow Function
12 const multiply = (a, b) => a * b;
13
14 // Function Calls
15 let x = 10, y = 3;
16
17 console.log("Addition using function declaration: " + add(x, y));
18 console.log("Subtraction using function expressions: " + subtract(x, y));
19 console.log("Multiplication using arrow functions: " + multiply(x, y));
20
```

The right-hand pane, titled "Output", displays the results of the console logs:

```
Addition using function declaration: 15
Subtraction using function expressions: 5
Multiplication using arrow function: 30
=== Code Execution Successful ===
```

The bottom of the image shows a Windows taskbar with the search bar, task view button, and several application icons including File Explorer, Edge, and VS Code. The system tray on the right indicates the date and time as "Wednesday 09:00" and the language as "ENG IN".

2.a) Program to implement a counter button using react class components

```
import React, { Component } from 'react';

class Counter extends Component {

  constructor(props) {

    super(props);

    this.state = {

      count: 0

    };

  }

  incrementCounter = () => {

    this.setState({

      count: this.state.count + 1

    });

  };

  render() {

    return (

      <div style={{ textAlign: 'center', marginTop: '50px' }}>

        <h1>React Counter Using Class Component</h1>

        <h2>Count: {this.state.count}</h2>

        <button onClick={this.incrementCounter}>Increment</button>

      </div>

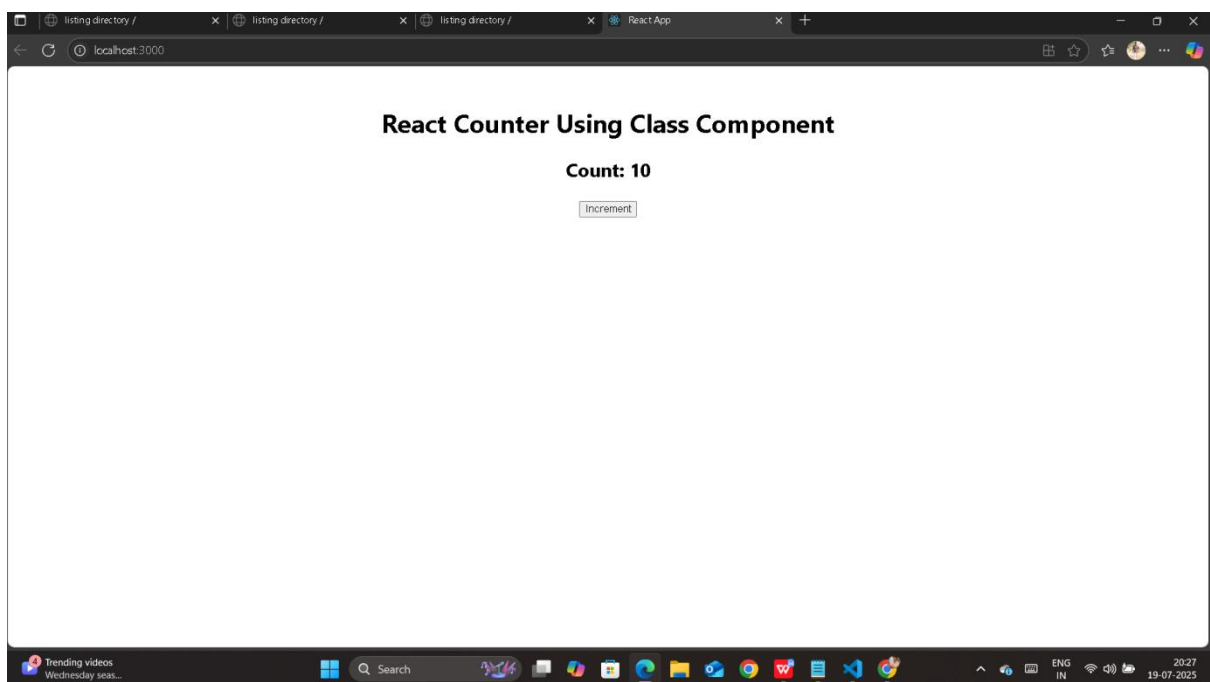
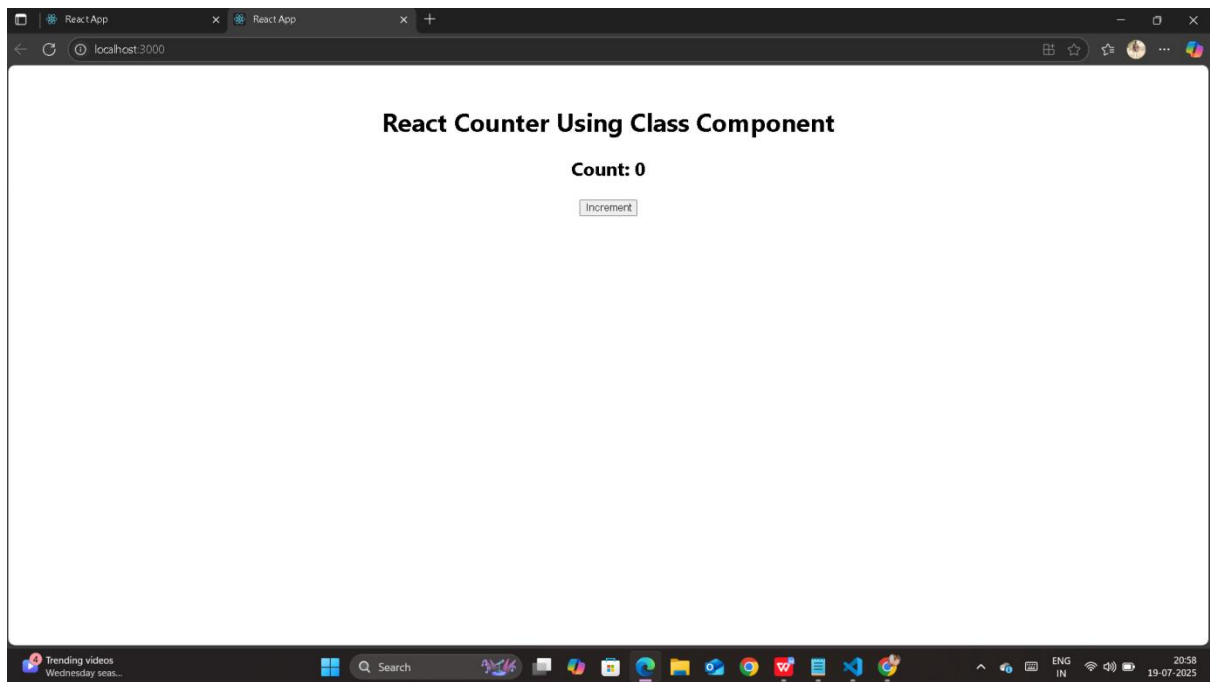
    );

  }

}

export default Counter;
```


Output:



2.b) Program to implement a counter button using react functional components

//Counter.js

```
import React, { useState } from 'react';

const Counter = () => {

  const [count, setCount] = useState(0);


  const handleIncrement = () => {

    setCount(count + 1);

  };


  return (

    <div style={{ textAlign: 'center', marginTop: '50px' }}>

      <h2>React Counter using Functional Component</h2>

      <p style={{ fontSize: '24px' }}>Count: {count}</p>

      <button onClick={handleIncrement} style={{

        padding: '10px 20px',

        fontSize: '18px',

        cursor: 'pointer',

        backgroundColor: '#4CAF50',

        color: 'white',

        border: 'none',

        borderRadius: '5px'

      }}>

        Increment

      </button>

    </div>

  );

};
```

```
export default Counter;
```

App.js

```
import React from 'react';
```

```
import Counter from './Counter';
```

```
function App() {
```

```
  return (
```

```
    <div className="App">
```

```
      <Counter />
```

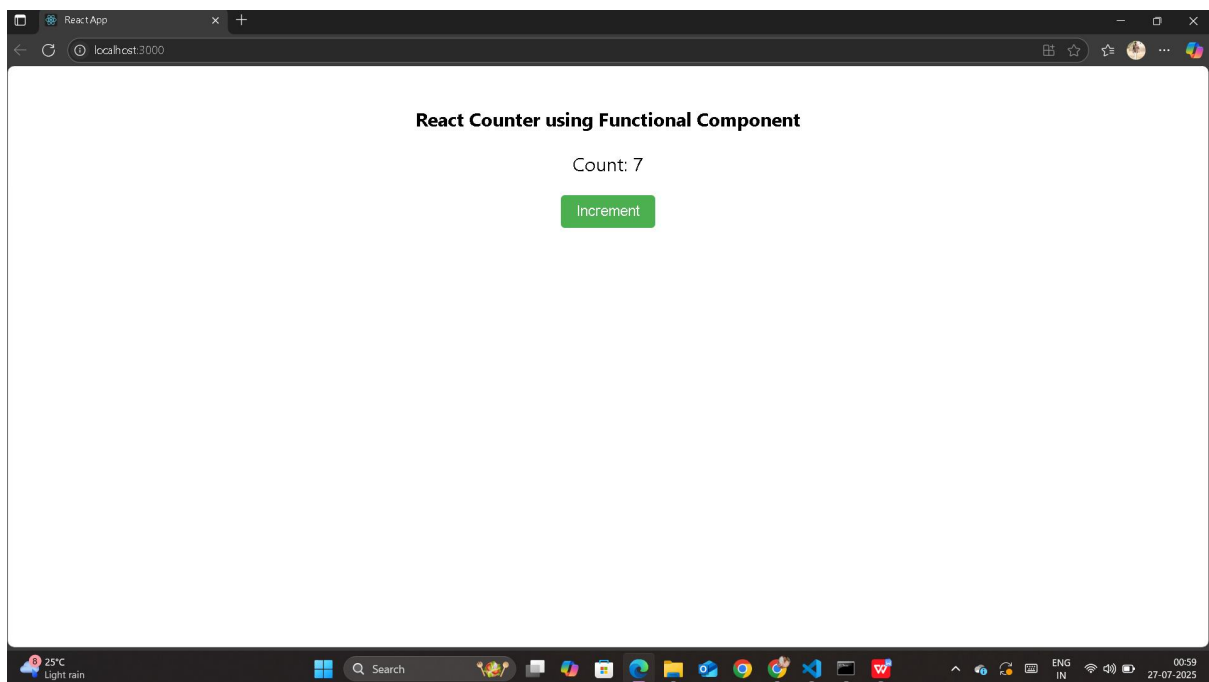
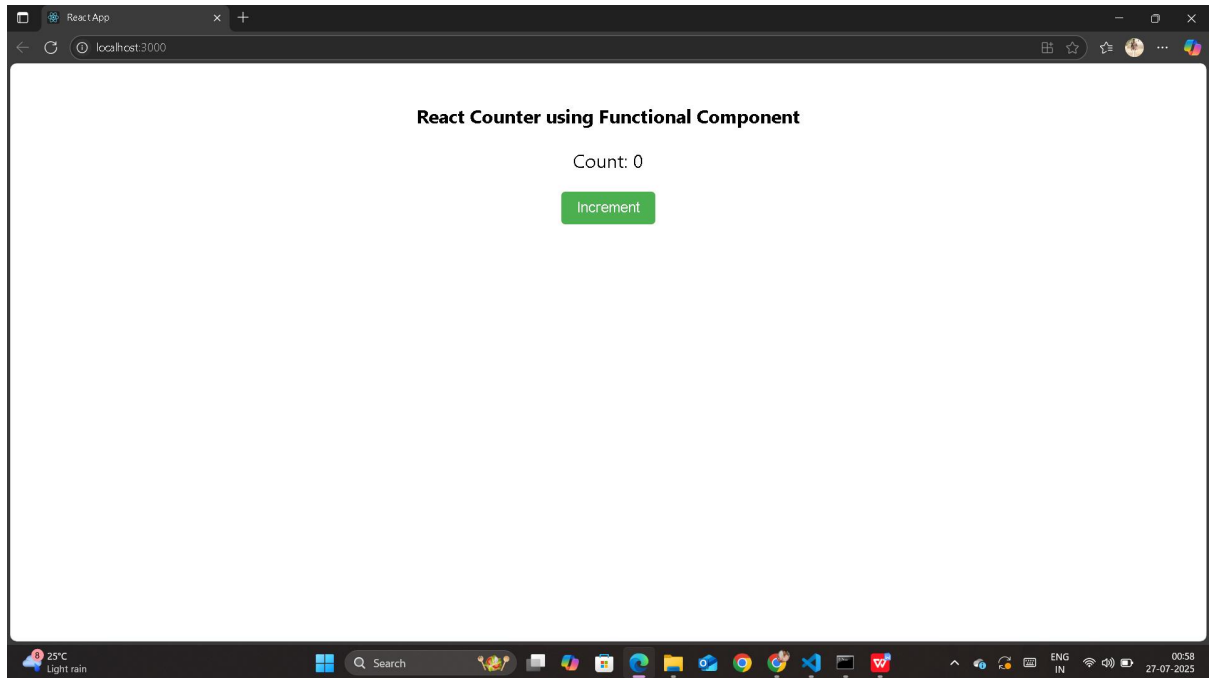
```
    </div>
```

```
  );
```

```
}
```

```
export default App;
```

Output:



2.c) Program to handle the button click events in functional components

//ButtonClickHandler.js

```
import React, { useState } from 'react';
```

```
const ButtonClickHandler = () => {
```

```
  const [message, setMessage] = useState('Click a button to see action');
```

```
  const handleGreet = () => {
```

```
    setMessage('Hello! You clicked Greet.');
```

```
  };
```

```
  const handleGoodbye = () => {
```

```
    setMessage('Goodbye! You clicked Exit.');
```

```
  };
```

```
  return (
```

```
    <div style={{ textAlign: 'center', marginTop: '50px' }}>
```

```
      <h2>React Button Click Event Handler</h2>
```

```
      <p style={{ fontSize: '20px' }}>{message}</p>
```

```
      <button onClick={handleGreet} style={buttonStyle}>Greet</button>
```

```
      <button   onClick={handleGoodbye}   style={{   ...buttonStyle,   backgroundColor: '#f44336' }}>Exit</button>
```

```
    </div>
```

```
  );
```

```
};
```

```
const buttonStyle = {
```

```
  margin: '10px',
```

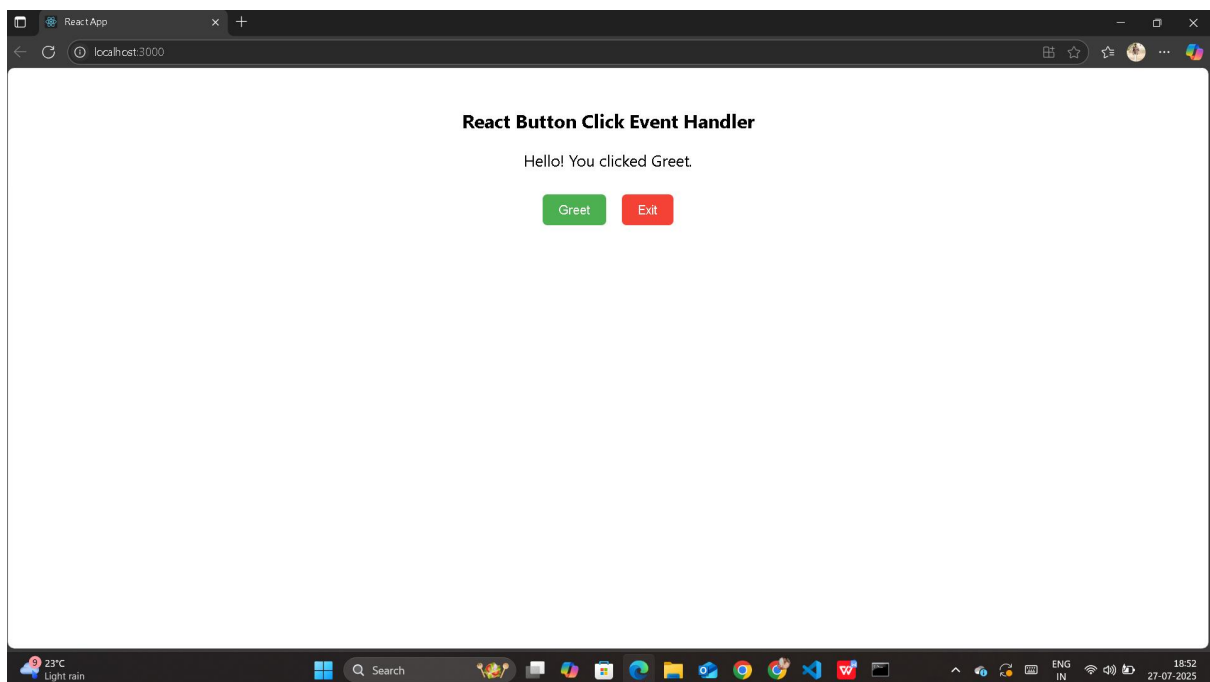
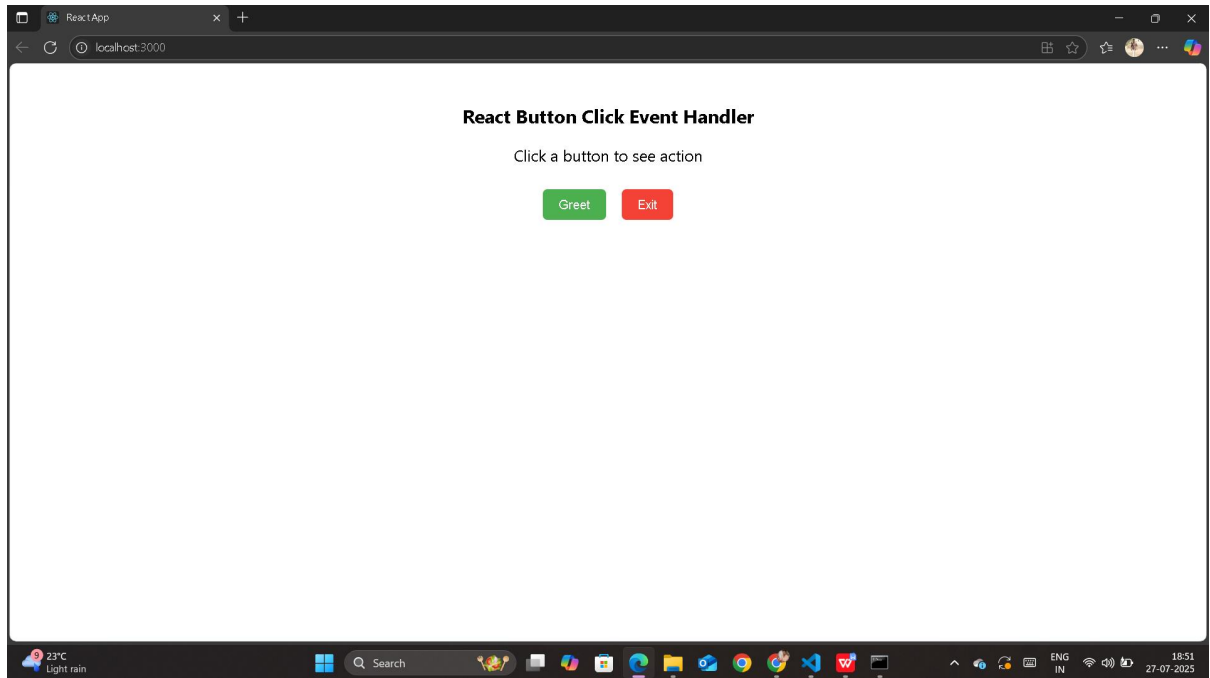
```
  padding: '10px 20px',
```

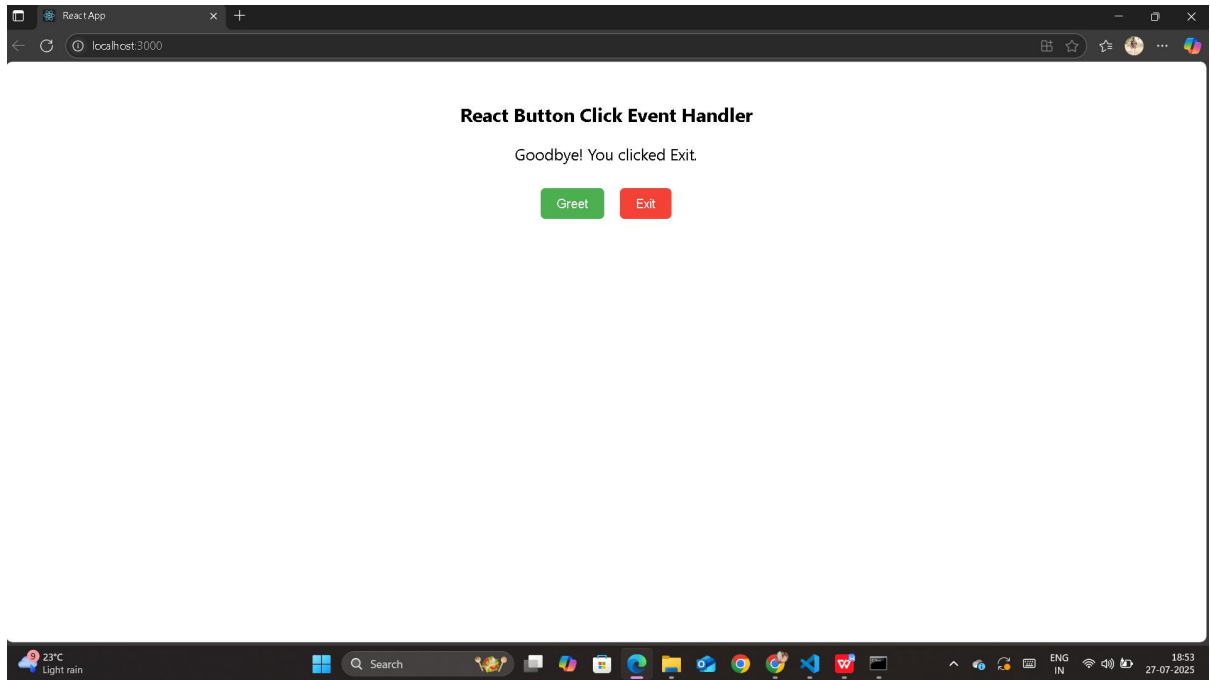
```
    fontSize: '16px',  
    borderRadius: '6px',  
    backgroundColor: '#4CAF50',  
    color: 'white',  
    border: 'none',  
    cursor: 'pointer'  
};  
  
export default ButtonClickHandler;
```

App.js

```
import React from 'react';  
import ButtonClickHandler from './ButtonClickHandler';  
  
function App() {  
  return (  
    <div className="App">  
      <ButtonClickHandler />  
    </div>  
  );  
}  
  
export default App;
```

Output:





2. d) Program to conditionally render a components in the browser

//App.js

```
import React, { useState } from 'react';

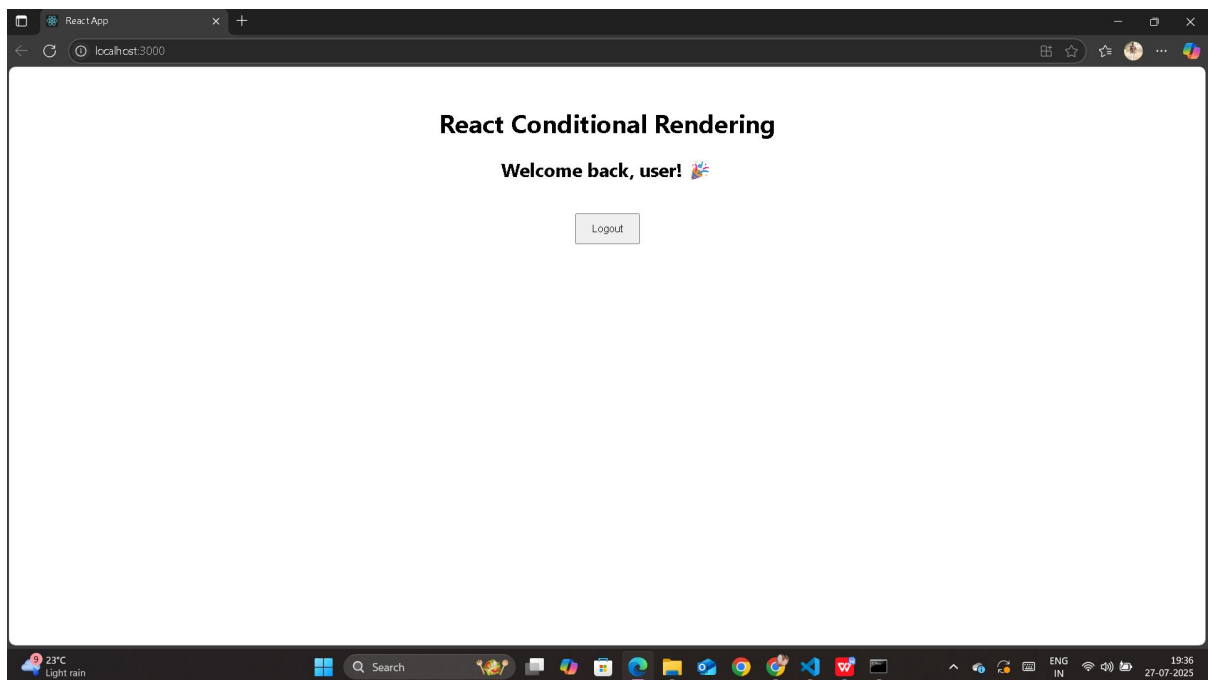
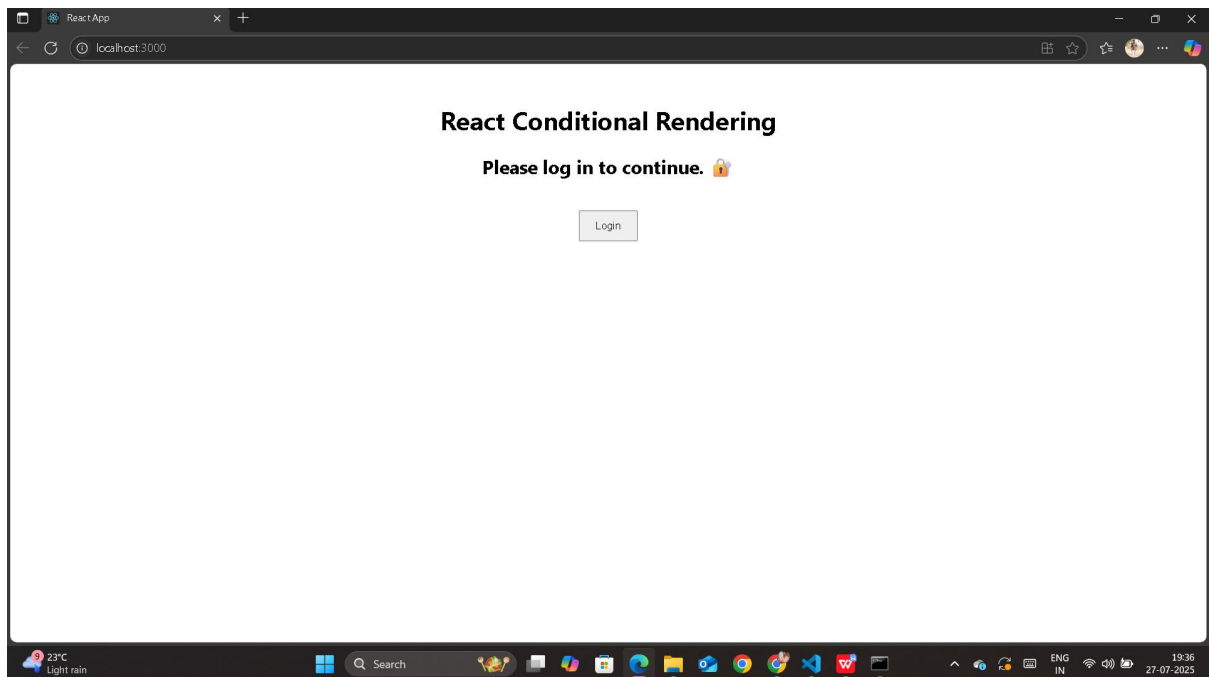
function Welcome() {
  return <h2>Welcome back, user! </h2>;
}

function PleaseLogin() {
  return <h2>Please log in to continue. </h2>;
}

function App() {
  const [isLoggedIn, setIsLoggedIn] = useState(false);
  const toggleLogin = () => {
    setIsLoggedIn(!isLoggedIn);
  };
  return (
    <div style={{ textAlign: 'center', marginTop: '50px' }}>
      <h1>React Conditional Rendering</h1>
      {isLoggedIn ? <Welcome /> : <PleaseLogin />}
      <button onClick={toggleLogin} style={{ marginTop: '20px', padding: '10px 20px' }}>
        {isLoggedIn ? 'Logout' : 'Login'}
      </button>
    </div>
  );
}

export default App;
```

Output:



2.e) Program to display text using String literals

// App.js

```
import React from 'react';
```

```
function App() {
```

```
  const name = "Sathish";
```

```
  const course = "ReactJS";
```

```
  // String literal using template literals
```

```
  const message = `Welcome ${name}! You are learning ${course}.`;
```

```
  return (
```

```
    <div>
```

```
      <h1>React String Literals Example</h1>
```

```
      <p>{message}</p>
```

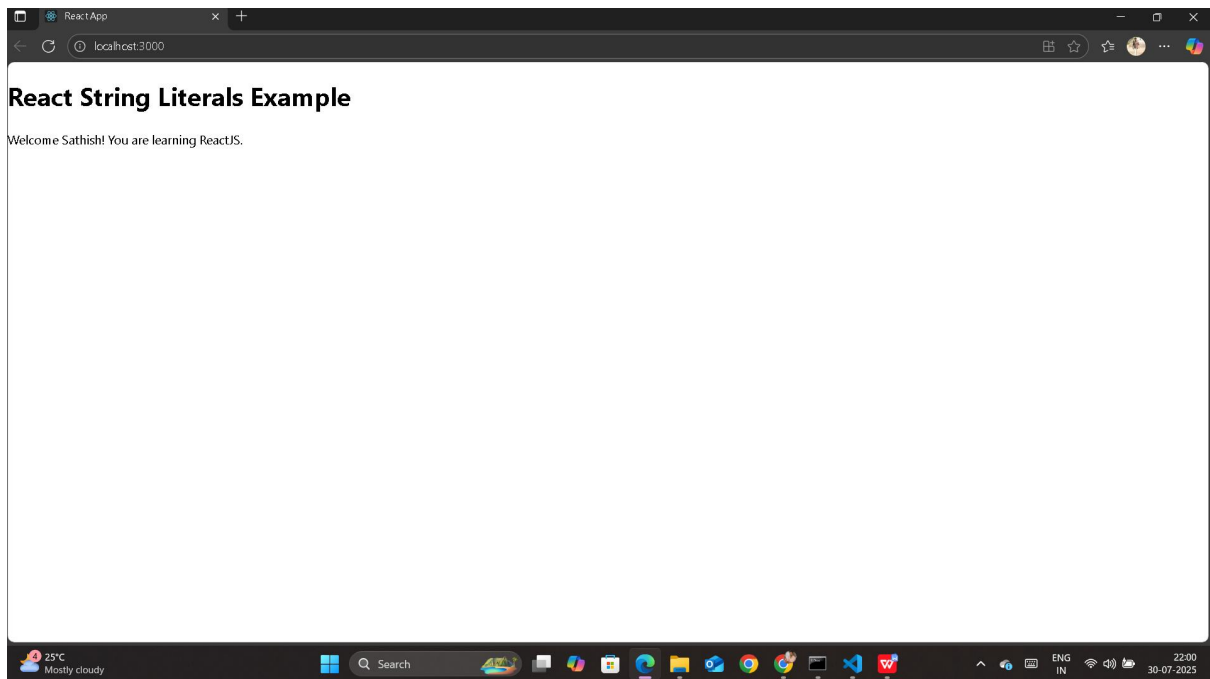
```
    </div>
```

```
  );
```

```
}
```

```
export default App;
```

Output:



3.a) Program to implement a counter button using React use State hook

App.js

```
import React, { useState } from 'react';
import './App.css'; // Optional if you want to add external styles

function App() {
  const [count, setCount] = useState(0);

  const increment = () => setCount(prev => prev + 1);
  const decrement = () => setCount(prev => prev - 1);
  const reset = () => setCount(0);

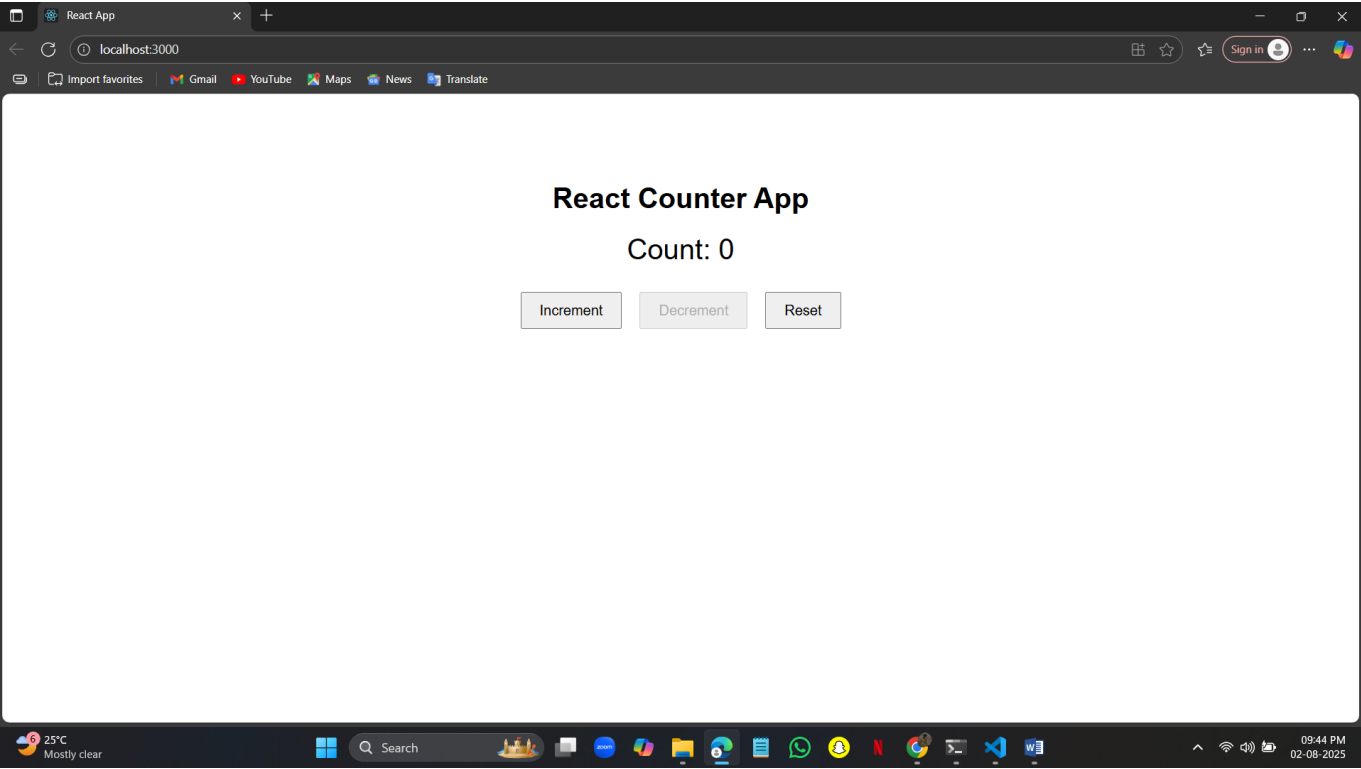
  const containerStyle = {
    textAlign: 'center',
    marginTop: '100px',
    fontFamily: 'Arial, sans-serif',
  };

  const buttonStyle = {
    margin: '10px',
    padding: '10px 20px',
    fontSize: '16px',
    cursor: 'pointer',
  };

  const countStyle = {
    fontSize: '32px',
    margin: '20px 0',
  };
}
```

```
return (  
  <div style={containerStyle}>  
    <h1>React Counter App</h1>  
    <div style={countStyle}>Count: {count}</div>  
    <button style={buttonStyle} onClick={increment}>Increment</button>  
    <button style={buttonStyle} onClick={decrement} disabled={count ===  
0}>Decrement</button>  
    <button style={buttonStyle} onClick={reset}>Reset</button>  
  </div>  
);  
}  
  
export default App;
```

Output:



3.b) Program to fetch the data from an API using React use Effect hook

App.js

```
import React, { useEffect, useState } from 'react';

function App() {
  const [users, setUsers] = useState([]);    // To store fetched user data
  const [loading, setLoading] = useState(true); // To manage loading state
  const [error, setError] = useState(null);   // To store error (if any)

  useEffect(() => {
    // Fetching data inside useEffect
    fetch('https://jsonplaceholder.typicode.com/users')
      .then(response => {
        if (!response.ok) {
          throw new Error('Network response was not ok');
        }
        return response.json(); // Parse JSON data
      })
      .then(data => {
        setUsers(data);
        setLoading(false);
      })
      .catch(error => {
        setError(error.message);
        setLoading(false);
      });
  }, []); // Empty dependency array means this runs once when component mounts

  return (
    <div style={{ padding: '30px', fontFamily: 'Arial' }}>
```



```
<h1>User List</h1>
```

```
{loading && <p>Loading...</p>}
```

```
{error && <p style={{ color: 'red' }}>Error: {error}</p>}
```

```
{!loading && !error && (
```

```
  <ul>
```

```
    {users.map(user => (
```

```
      <li key={user.id}>
```

```
        <strong>{user.name}</strong> — {user.email}
```

```
      </li>
```

```
    )}
```

```
  </ul>
```

```
)}
```

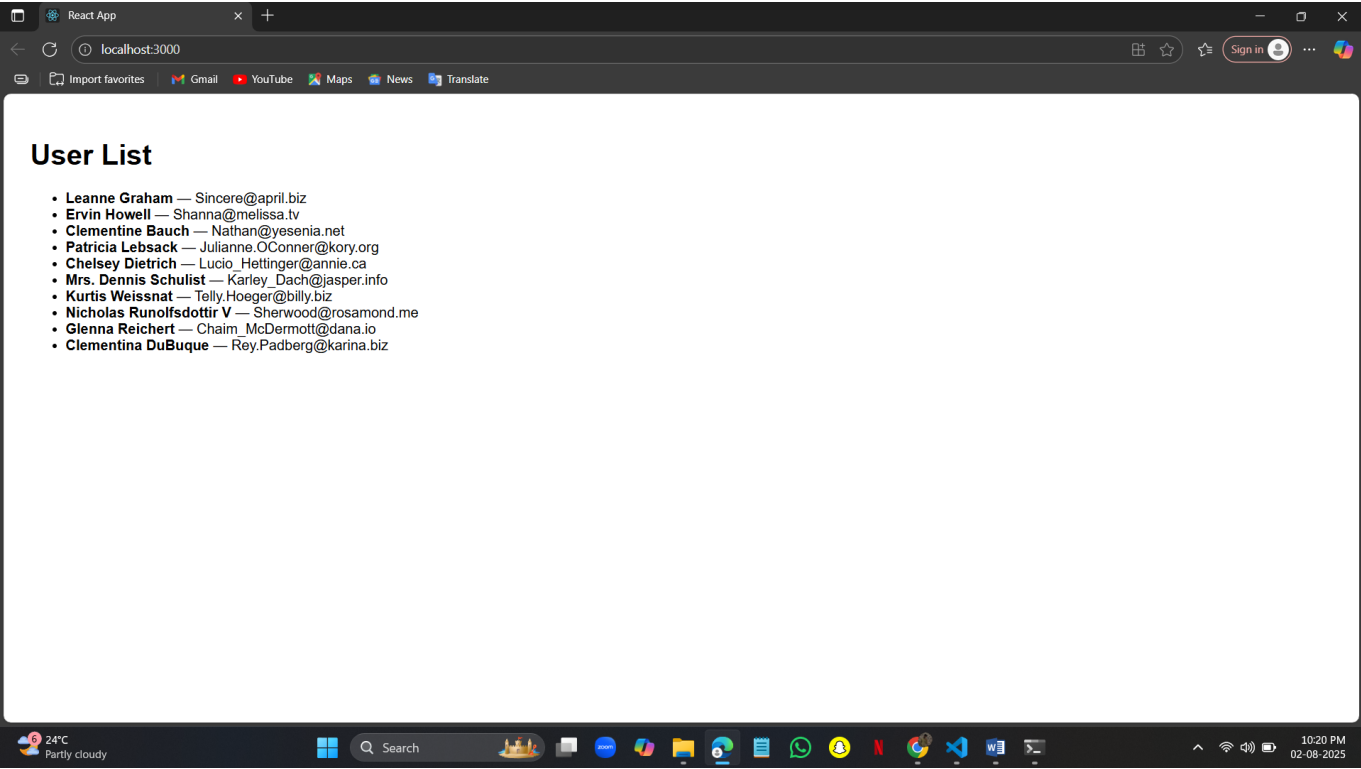
```
</div>
```

```
);
```

```
}
```

```
export default App;
```

Output:



3.c) Program with two react components sharing data using Props

App.js

```
import React from 'react';
import UserProfile from './UserProfile';
import Greeting from './Greeting';

function App() {
  const user = {
    name: 'Alice',
    age: 28,
    location: 'New York'
  };

  return (
    <div>
      <h1>React Props Sharing Example</h1>
      <Greeting name={user.name} />
      <UserProfile user={user} />
    </div>
  );
}

export default App;
```

Greeting.js

```
import React from 'react';

function Greeting(props) {
  return <h2>Hello, {props.name}! 🖐️</h2>;
}
```

```
export default Greeting;
```

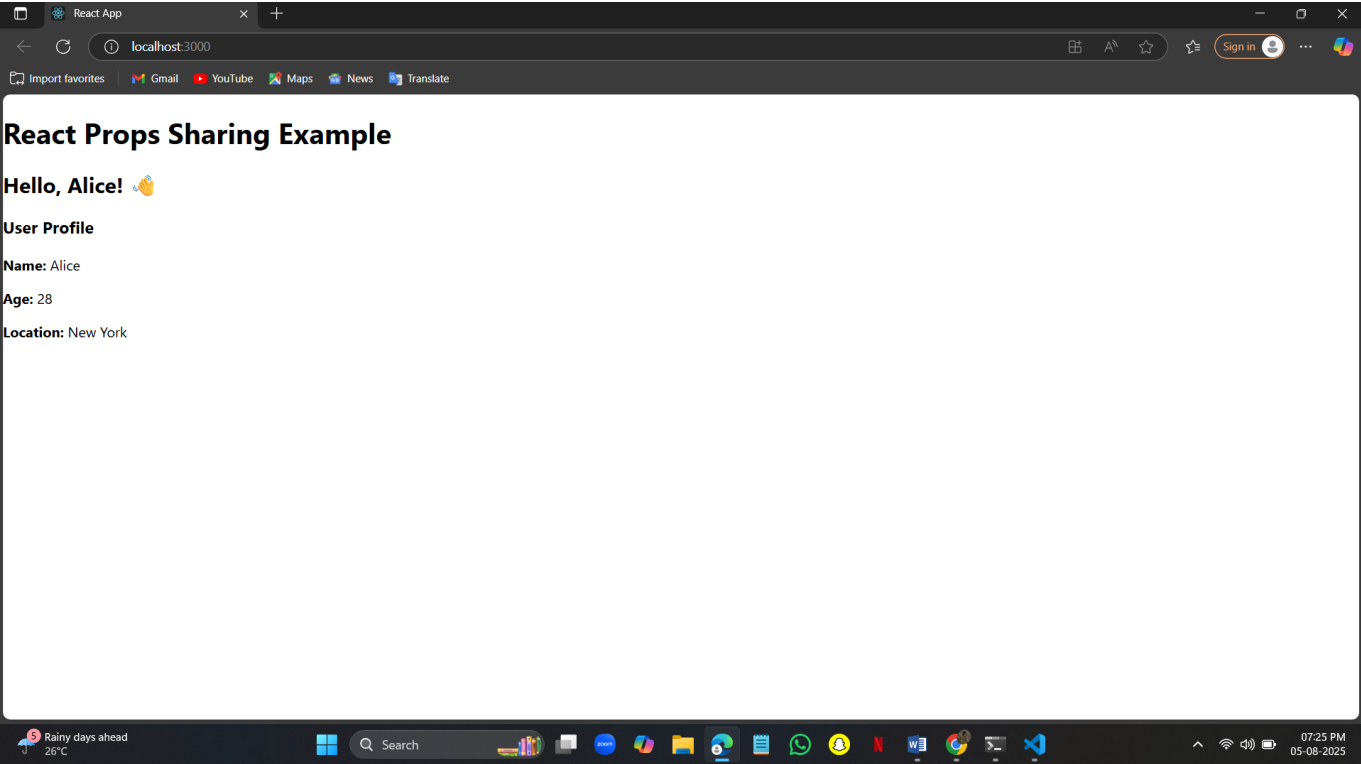
UserProfile.js

```
import React from 'react';
```

```
function UserProfile(props) {  
  const { name, age, location } = props.user;  
  
  return (  
    <div>  
      <h3>User Profile</h3>  
      <p><strong>Name:</strong> {name}</p>  
      <p><strong>Age:</strong> {age}</p>  
      <p><strong>Location:</strong> {location}</p>  
    </div>  
  );  
}
```

```
export default UserProfile;
```

Output:



3.d) Program to implement the forms in react

App.js

```
import React, { useState } from 'react';

function App() {
  // Step 1: Define state for form inputs and submitted data
  const [formData, setFormData] = useState({
    name: "",
    email: ""
  });

  const [submittedData, setSubmittedData] = useState(null);

  // Step 2: Handle input changes
  const handleChange = (event) => {
    const { name, value } = event.target;
    setFormData((prevData) => ({
      ...prevData,
      [name]: value
    }));
  };

  // Step 3: Handle form submission
  const handleSubmit = (event) => {
    event.preventDefault(); // prevent page reload
    setSubmittedData(formData); // store submitted data
    setFormData({ name: "", email: "" }); // reset form
  };

  return (
```

```
<div style={{ padding: '20px' }}>
```

```
<h1>React Form Example</h1>
```

```
{/* Step 4: Form UI */}
```

```
<form onSubmit={handleSubmit}>
```

```
<div>
```

```
<label>Name: </label>
```

```
<input
```

```
  type="text"
```

```
  name="name"
```

```
  value={formData.name}
```

```
  onChange={handleChange}
```

```
  required
```

```
</div>
```

```
<br />
```

```
<div>
```

```
<label>Email: </label>
```

```
<input
```

```
  type="email"
```

```
  name="email"
```

```
  value={formData.email}
```

```
  onChange={handleChange}
```

```
  required
```

```
</div>
```

```
<br />
```

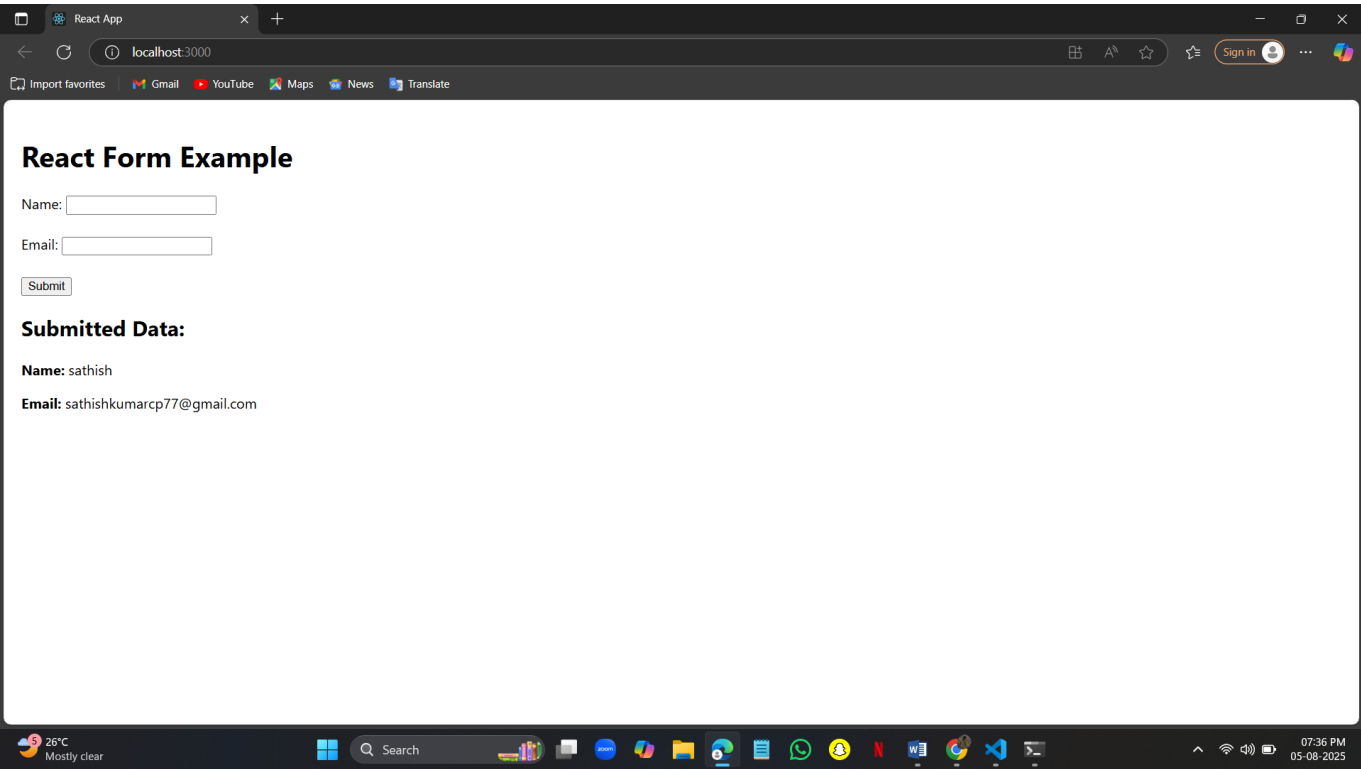
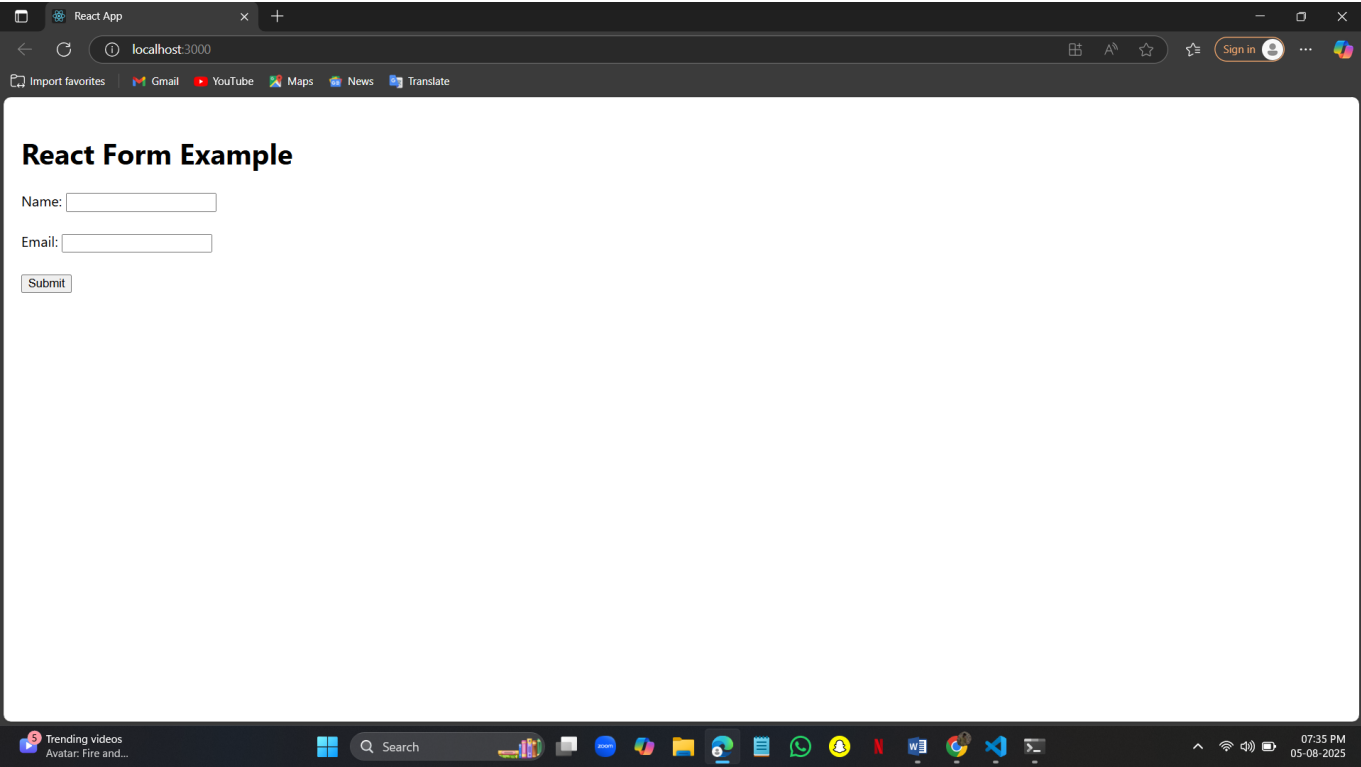
```
<button type="submit">Submit</button>
```

```
</form>
```

```
{/* Step 5: Display submitted data */}
{submittedData && (
  <div style={{ marginTop: '20px' }}>
    <h2>Submitted Data:</h2>
    <p><strong>Name:</strong> {submittedData.name}</p>
    <p><strong>Email:</strong> {submittedData.email}</p>
  </div>
)}
</div>
);
}

export default App;
```


Output:



3.e) Program to implement the iterative rendering using map() function.

App.js

```
import React from 'react';

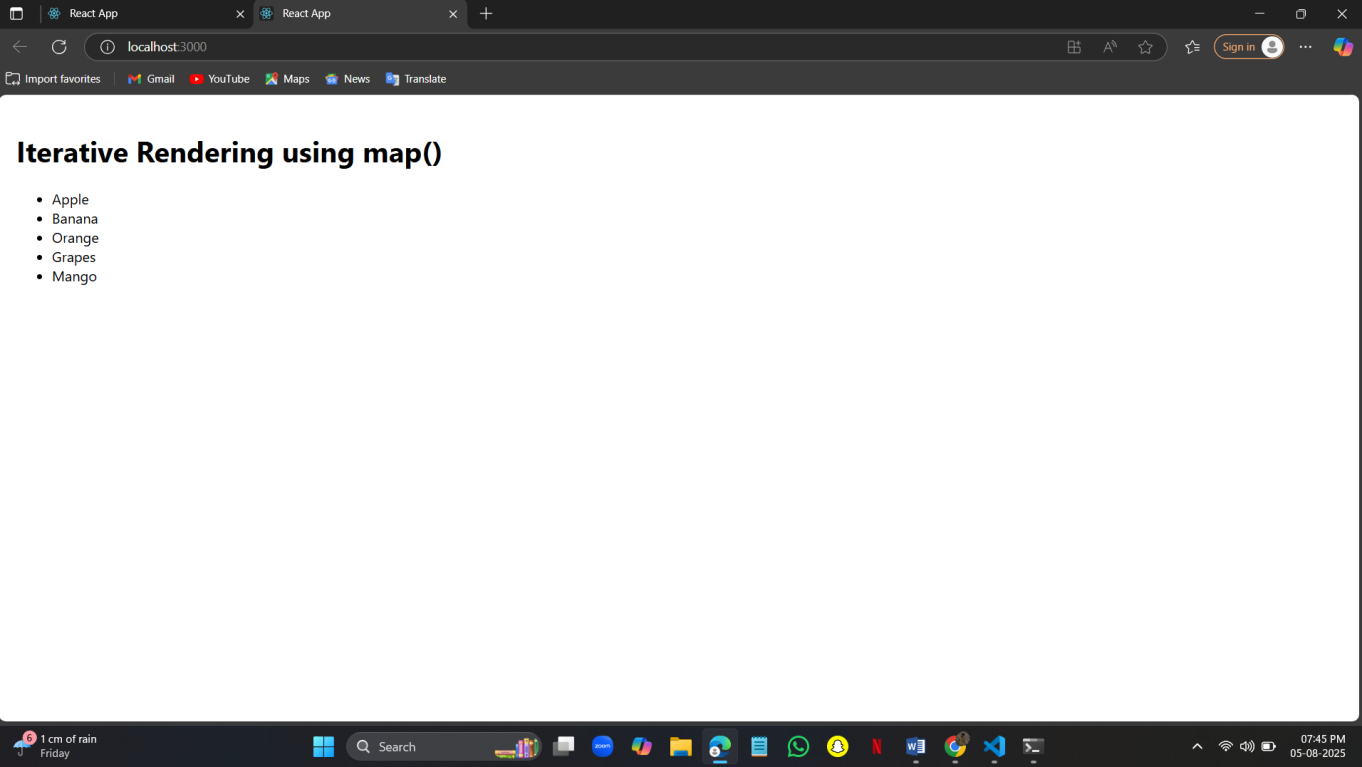
function App() {
  // Array of items to render
  const fruits = ['Apple', 'Banana', 'Orange', 'Grapes', 'Mango'];

  return (
    <div style={{ padding: '20px' }}>
      <h1>Iterative Rendering using map()</h1>

      <ul>
        { /* Using map() to render list items */}
        {fruits.map((fruit, index) => (
          <li key={index}>{fruit}</li>
        ))}
      </ul>
    </div>
  );
}

export default App;
```

Output:



4. Introduction to Git and GitHub

a. Setup

Program:

a. Install Git on Local Machine

For Windows:

1. Go to: <https://git-scm.com/download/win>
2. Download the installer.
3. Run the setup with default options.
4. Verify installation:
5. `git --version`

For macOS:

```
brew install git git
--version
```

For Linux (Debian/Ubuntu):

```
sudo apt update sudo
apt install git
git --version
```

b. Configure Git (Username and Email)

```
git config --global user.name "Your Name"
git config --global user.email "youremail@example.com"
```

To verify the configuration: `git`

```
config --global --list
```

c. Create GitHub Account and Generate Personal Access Token

Step 1: Create GitHub Account

1. Go to: <https://github.com/>
2. Click on **Sign up**.
3. Enter your email, password, and username.
4. Verify email and complete setup.

Step 2: Generate Personal Access Token (PAT)

1. Log in to GitHub.
2. Click your profile icon → **Settings** → **Developer settings** → **Tokens (classic)** → **Generate new token (classic)**.
3. Set expiration (recommended: 90 days).

4. Select scopes like:
 - `repo` (for full control of private repositories)
 - `workflow` ○ `read:org`
5. Click **Generate token**.
6. **Copy the token** (you won't see it again).

Output:

```
Command Prompt
Microsoft Windows [Version 10.0.19045.5371]
(c) Microsoft Corporation. All rights reserved.

C:\Users\KEC>git --version
git version 2.50.1.windows.1

C:\Users\KEC>
```

```
Command Prompt
Microsoft Windows [Version 10.0.19045.5371]
(c) Microsoft Corporation. All rights reserved.

C:\Users\KEC>git --version
git version 2.50.1.windows.1

C:\Users\KEC>git config --global user.name "Your Name"

C:\Users\KEC>git config --global user.email "youremail@example.com"

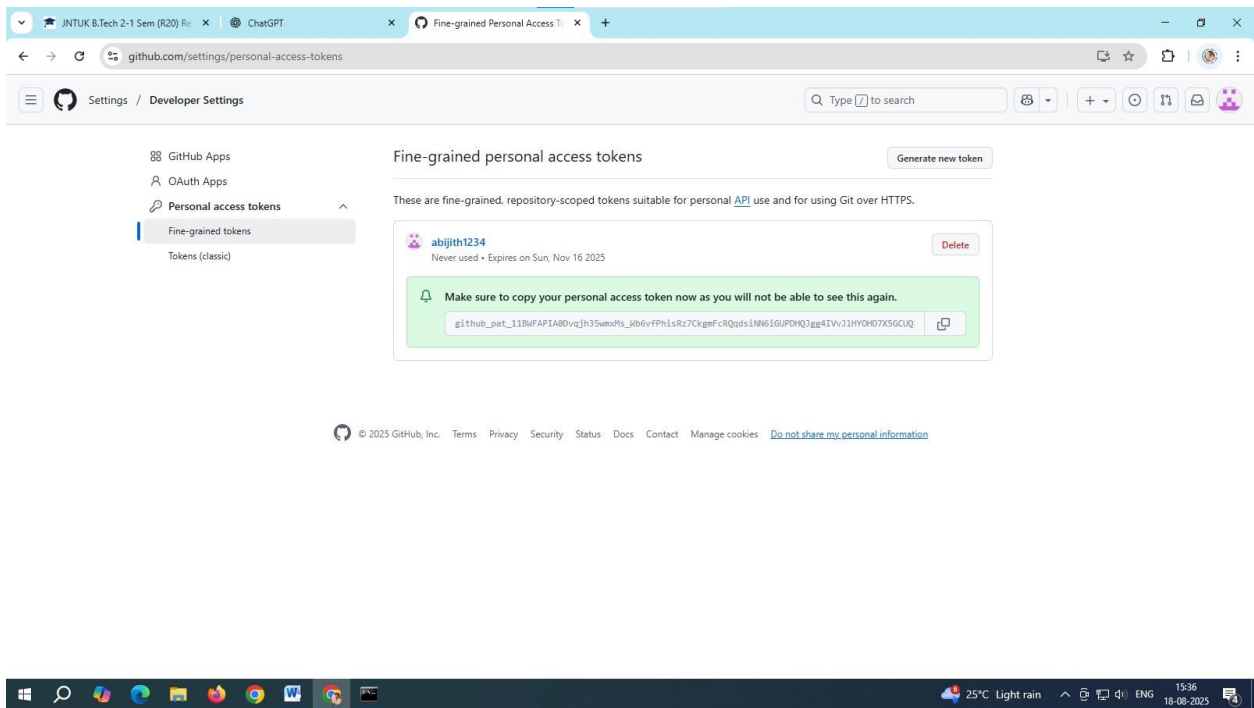
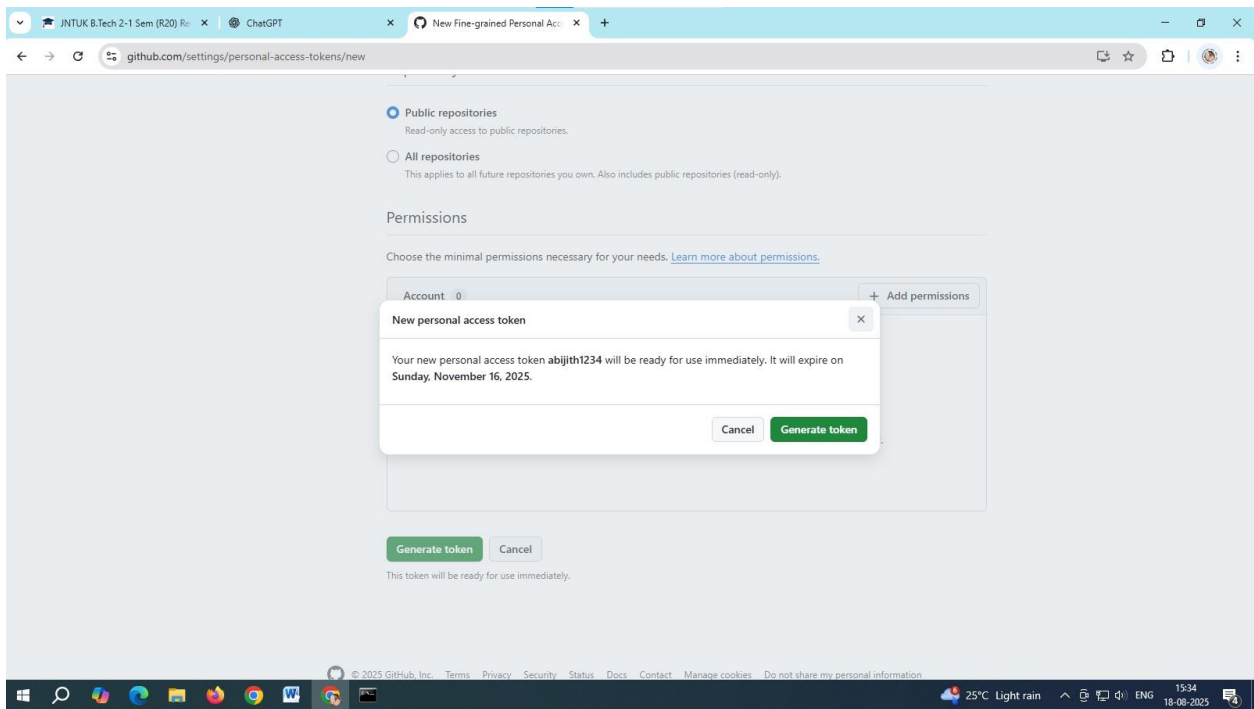
C:\Users\KEC>git config --global user.sathish "sathish"

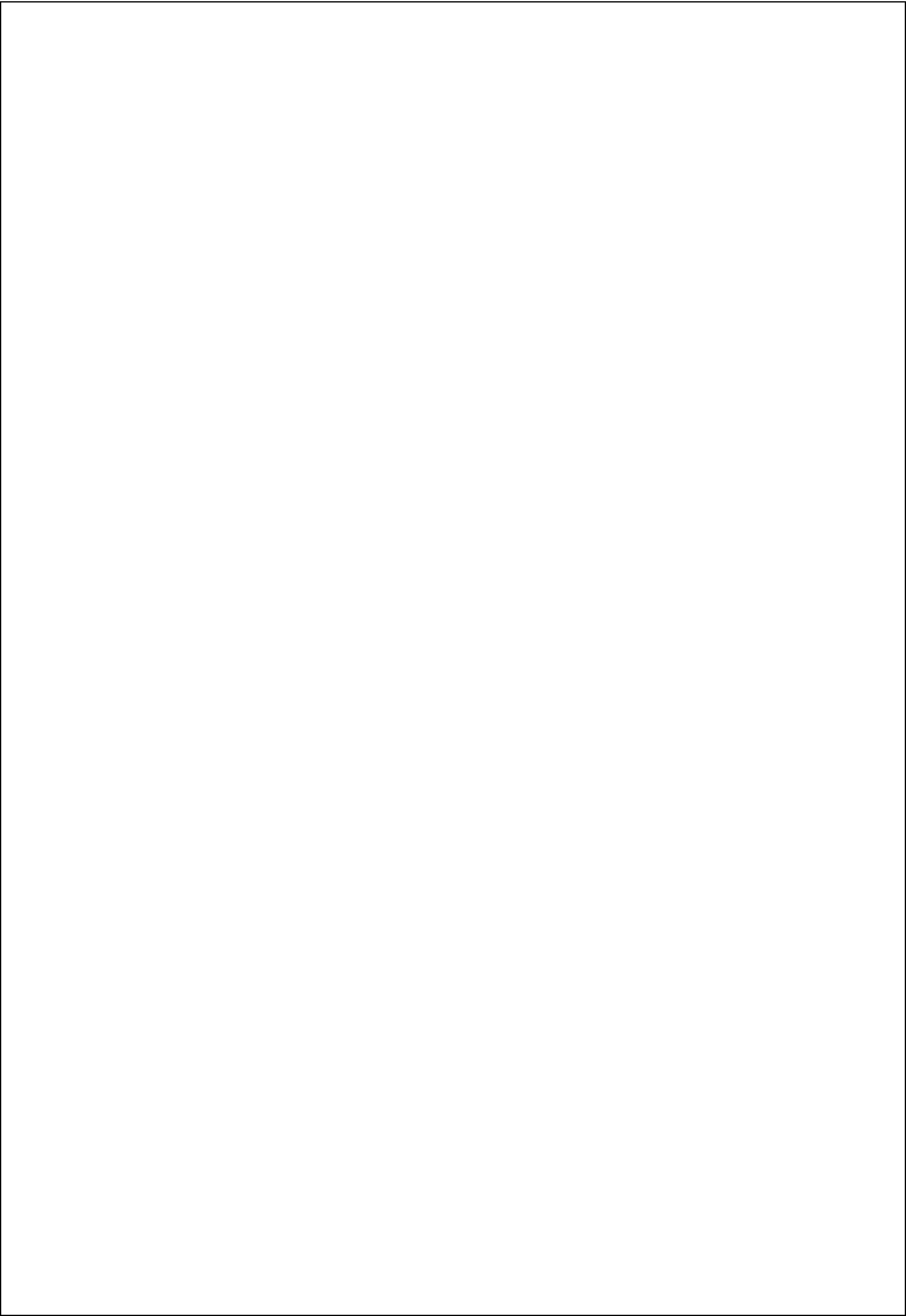
C:\Users\KEC>git config --global user.name "sathishkumar"

C:\Users\KEC>git config --global user.email "sathishkumarc77@gmail.com"

C:\Users\KEC>git config --global --list
user.name=sathishkumar
user.email=sathishkumarc77@gmail.com
user.sathish=sathish

C:\Users\KEC>
```





4 (b) Basic Git Workflow

Program:

Step 1: Initialize git

```
git init
```

Step 2: Create a file

```
echo "Hello Git" > file1.txt
```

Step 3: Add the file

```
git add file1.txt
```

Step 4: Commit the file

```
git commit -m "Initial commit"
```

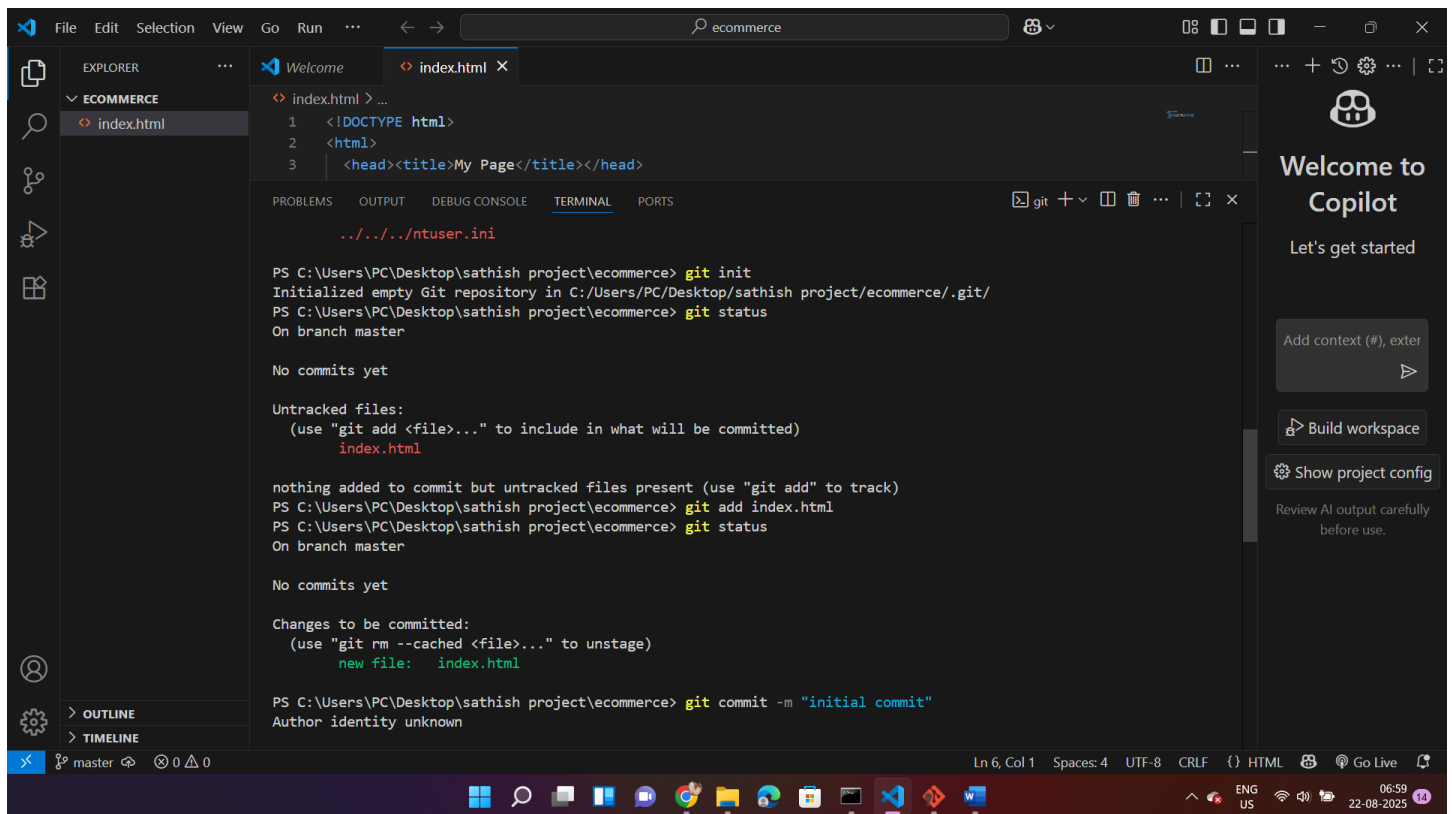
Step 5: Connect with GitHub remote

```
git remote add origin https://github.com/username/repo.git
```

Step 6: Push to remote repository

```
git push -u origin main
```

Output:



The screenshot shows the Visual Studio Code interface with a terminal window open. The terminal displays the output of several git commands executed in a PowerShell prompt. The commands and their outputs are as follows:

```
PS C:\Users\PC\Desktop\sathish project\ecommerce> git init
Initialized empty Git repository in C:/Users/PC/Desktop/sathish project/ecommerce/.git/
PS C:\Users\PC\Desktop\sathish project\ecommerce> git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        index.html

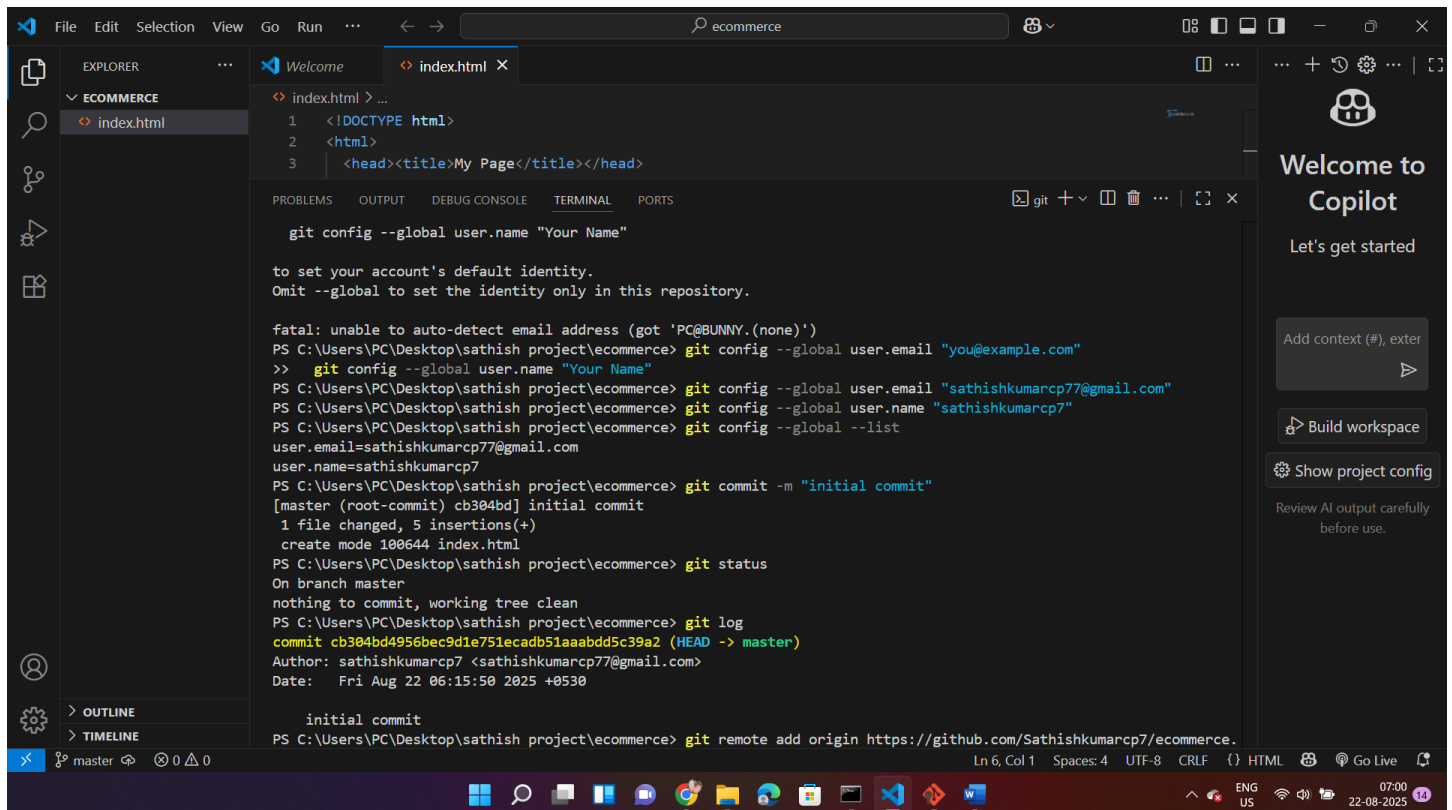
nothing added to commit but untracked files present (use "git add" to track)
PS C:\Users\PC\Desktop\sathish project\ecommerce> git add index.html
PS C:\Users\PC\Desktop\sathish project\ecommerce> git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   index.html

PS C:\Users\PC\Desktop\sathish project\ecommerce> git commit -m "initial commit"
Author identity unknown
```

The Explorer sidebar on the left shows a folder named 'ECOMMERCE' containing a file 'index.html'. The editor window shows the content of 'index.html', which is a basic HTML document with a doctype, html tag, and a head section with a title 'My Page'.



The screenshot shows the Visual Studio Code interface with a terminal window open. The terminal displays the output of several git commands executed in a PowerShell prompt. The commands and their outputs are as follows:

```
git config --global user.name "Your Name"

to set your account's default identity.
Omit --global to set the identity only in this repository.

fatal: unable to auto-detect email address (got 'PC@BUNNY.(none)')
PS C:\Users\PC\Desktop\sathish project\ecommerce> git config --global user.email "you@example.com"
>> git config --global user.name "Your Name"
PS C:\Users\PC\Desktop\sathish project\ecommerce> git config --global user.email "sathishkumarp77@gmail.com"
PS C:\Users\PC\Desktop\sathish project\ecommerce> git config --global user.name "sathishkumarp7"
PS C:\Users\PC\Desktop\sathish project\ecommerce> git config --global --list
user.email=sathishkumarp77@gmail.com
user.name=sathishkumarp7
PS C:\Users\PC\Desktop\sathish project\ecommerce> git commit -m "initial commit"
[master (root-commit) cb304bd] initial commit
1 file changed, 5 insertions(+)
create mode 100644 index.html
PS C:\Users\PC\Desktop\sathish project\ecommerce> git status
On branch master
nothing to commit, working tree clean
PS C:\Users\PC\Desktop\sathish project\ecommerce> git log
commit cb304bd4956bec9d1e751ecadb51aaabdd5c39a2 (HEAD -> master)
Author: sathishkumarp7 <sathishkumarp77@gmail.com>
Date: Fri Aug 22 06:15:50 2025 +0530

    initial commit

PS C:\Users\PC\Desktop\sathish project\ecommerce> git remote add origin https://github.com/Sathishkumarp7/ecommerce.
```

The Explorer sidebar on the left shows a folder named 'ECOMMERCE' containing a file 'index.html'. The editor window shows the content of 'index.html', which is a basic HTML document with a doctype, html tag, and a head section with a title 'My Page'.



4(c) Branching and Collaboration in Git

PROGRAM:

Step 1: Initialize repository

```
git init demo-repo
```

```
cd demo-repo
```

Step 2: Create a file and commit in main

```
echo "Hello from main branch" > file.txt
```

```
git add file.txt
```

```
git commit -m "Initial commit on main"
```

Step 3: Create a new branch

```
git checkout -b feature1
```

Step 4: Edit file in feature1 branch

```
echo "Hello from feature1 branch" > file.txt
```

```
git add file.txt
```

```
git commit -m "Update file in feature1 branch"
```

Step 5: Switch back to main branch

```
git checkout main
```

Step 6: Edit file in main branch

```
echo "Hello from main branch - modified" > file.txt
```

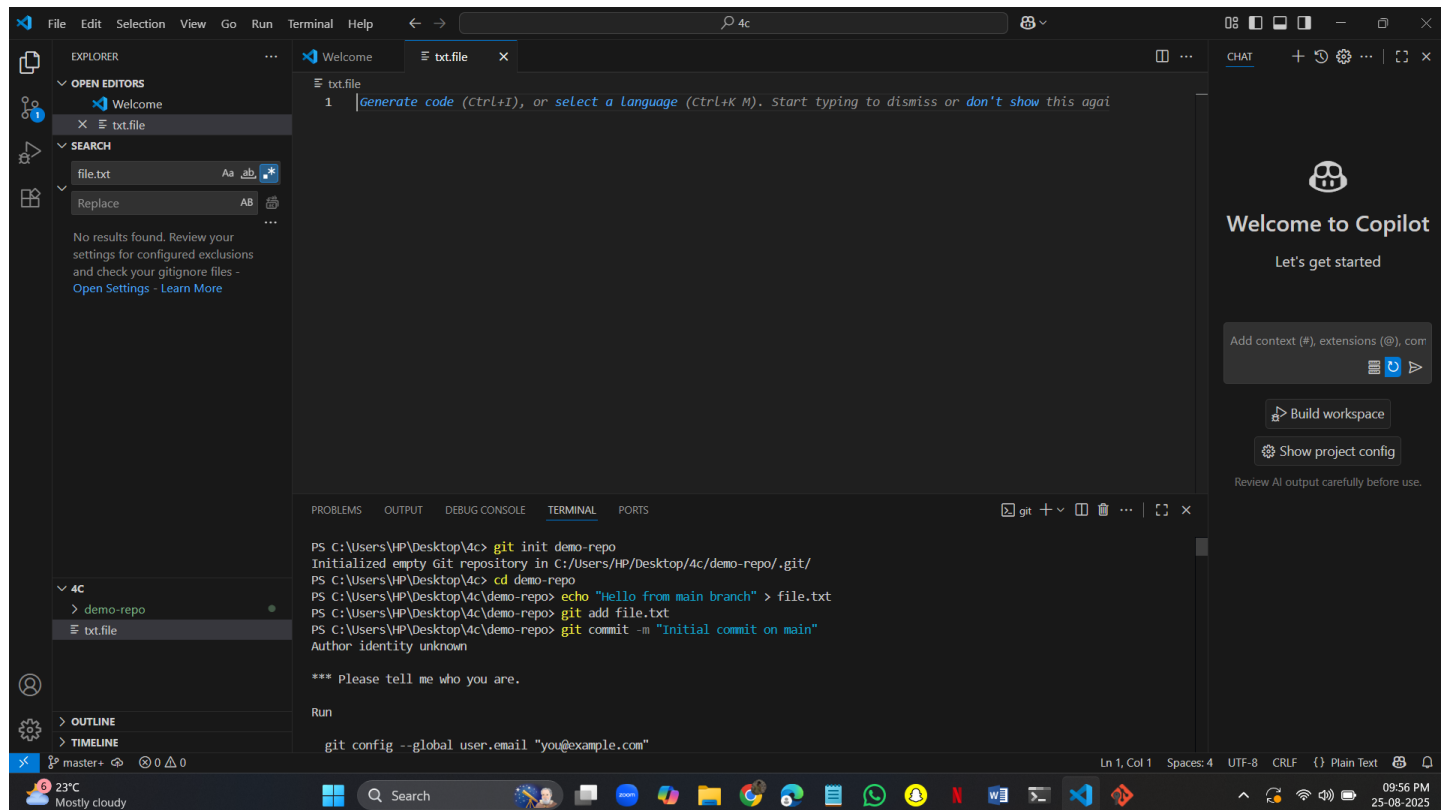
```
git add file.txt
```

```
git commit -m "Update file in main branch"
```

Step 7: Merge feature1 into main

```
git merge feature1
```

OUTPUT:



The screenshot shows the Visual Studio Code (VS Code) interface. The Explorer sidebar on the left shows a file named 'txt.file'. The main editor area displays the content of 'txt.file', which is a prompt from Copilot: 'Generate code (Ctrl+I), or select a language (Ctrl+K M). Start typing to dismiss or don't show this again'. The bottom panel shows the TERMINAL view with the following output:

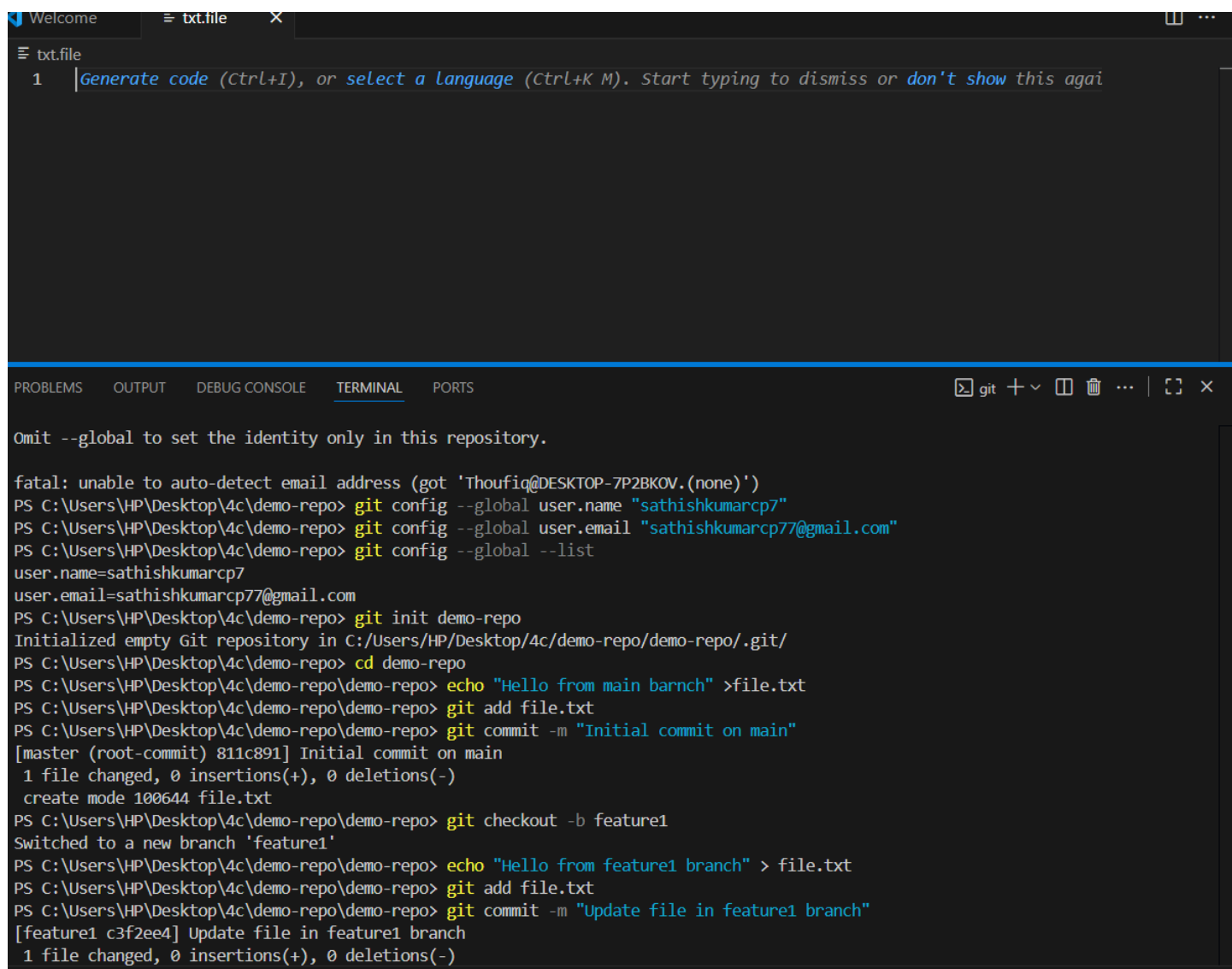
```
PS C:\Users\HP\Desktop\4c> git init demo-repo
Initialized empty Git repository in C:/Users/HP/Desktop/4c/demo-repo/.git/
PS C:\Users\HP\Desktop\4c> cd demo-repo
PS C:\Users\HP\Desktop\4c\demo-repo> echo "Hello from main branch" > file.txt
PS C:\Users\HP\Desktop\4c\demo-repo> git add file.txt
PS C:\Users\HP\Desktop\4c\demo-repo> git commit -m "Initial commit on main"
Author identity unknown

*** Please tell me who you are.

Run

git config --global user.email "you@example.com"
```

The right sidebar shows the 'Welcome to Copilot' panel with options to 'Build workspace' and 'Show project config'.



The screenshot shows the Visual Studio Code (VS Code) interface with the TERMINAL view selected. The terminal displays the following output:

```
Omit --global to set the identity only in this repository.

fatal: unable to auto-detect email address (got 'Thoufiq@DESKTOP-7P2BKOV.(none)')
PS C:\Users\HP\Desktop\4c\demo-repo> git config --global user.name "sathishkumarc7"
PS C:\Users\HP\Desktop\4c\demo-repo> git config --global user.email "sathishkumarc77@gmail.com"
PS C:\Users\HP\Desktop\4c\demo-repo> git config --global --list
user.name=sathishkumarc7
user.email=sathishkumarc77@gmail.com
PS C:\Users\HP\Desktop\4c\demo-repo> git init demo-repo
Initialized empty Git repository in C:/Users/HP/Desktop/4c/demo-repo/demo-repo/.git/
PS C:\Users\HP\Desktop\4c\demo-repo> cd demo-repo
PS C:\Users\HP\Desktop\4c\demo-repo\demo-repo> echo "Hello from main barnch" >file.txt
PS C:\Users\HP\Desktop\4c\demo-repo\demo-repo> git add file.txt
PS C:\Users\HP\Desktop\4c\demo-repo\demo-repo> git commit -m "Initial commit on main"
[master (root-commit) 811c891] Initial commit on main
 1 file changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 file.txt
PS C:\Users\HP\Desktop\4c\demo-repo> git checkout -b feature1
Switched to a new branch 'feature1'
PS C:\Users\HP\Desktop\4c\demo-repo\demo-repo> echo "Hello from feature1 branch" > file.txt
PS C:\Users\HP\Desktop\4c\demo-repo\demo-repo> git add file.txt
PS C:\Users\HP\Desktop\4c\demo-repo\demo-repo> git commit -m "Update file in feature1 branch"
[feature1 c3f2ee4] Update file in feature1 branch
 1 file changed, 0 insertions(+), 0 deletions(-)
```

```
txt.file
1 Generate code (Ctrl+I), or select a language (Ctrl+K M). Start typing to dismiss or don't show this again

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS git + - [ ] [ ] ... | [ ] [ ] x

PS C:\Users\HP\Desktop\4c\demo-repo\demo-repo> git checkout master
Switched to branch 'master'
PS C:\Users\HP\Desktop\4c\demo-repo\demo-repo> echo "Hello from master branch - modifier">file.txt
error: pathspec 'main' did not match any file(s) known to git
PS C:\Users\HP\Desktop\4c\demo-repo\demo-repo> git checkout master
Switched to branch 'master'
PS C:\Users\HP\Desktop\4c\demo-repo\demo-repo> echo "Hello from master branch - modifier">file.txt
PS C:\Users\HP\Desktop\4c\demo-repo\demo-repo> git add file.txt
PS C:\Users\HP\Desktop\4c\demo-repo\demo-repo> git commit -m "Update file in main branch"
[master 81089ed] Update file in main branch
error: pathspec 'main' did not match any file(s) known to git
PS C:\Users\HP\Desktop\4c\demo-repo\demo-repo> git checkout master
Switched to branch 'master'
PS C:\Users\HP\Desktop\4c\demo-repo\demo-repo> echo "Hello from master branch - modifier">file.txt
PS C:\Users\HP\Desktop\4c\demo-repo\demo-repo> git add file.txt
PS C:\Users\HP\Desktop\4c\demo-repo\demo-repo> git commit -m "Update file in main branch"
error: pathspec 'main' did not match any file(s) known to git
PS C:\Users\HP\Desktop\4c\demo-repo\demo-repo> git checkout master
Switched to branch 'master'
PS C:\Users\HP\Desktop\4c\demo-repo\demo-repo> echo "Hello from master branch - modifier">file.txt
PS C:\Users\HP\Desktop\4c\demo-repo\demo-repo> git add file.txt
error: pathspec 'main' did not match any file(s) known to git
PS C:\Users\HP\Desktop\4c\demo-repo\demo-repo> git checkout master
Switched to branch 'master'
```

Ln 1, Col 1 Spaces: 4

PORTS

git + v | ... | x

Ln 1 Col 1 Spaces: 4

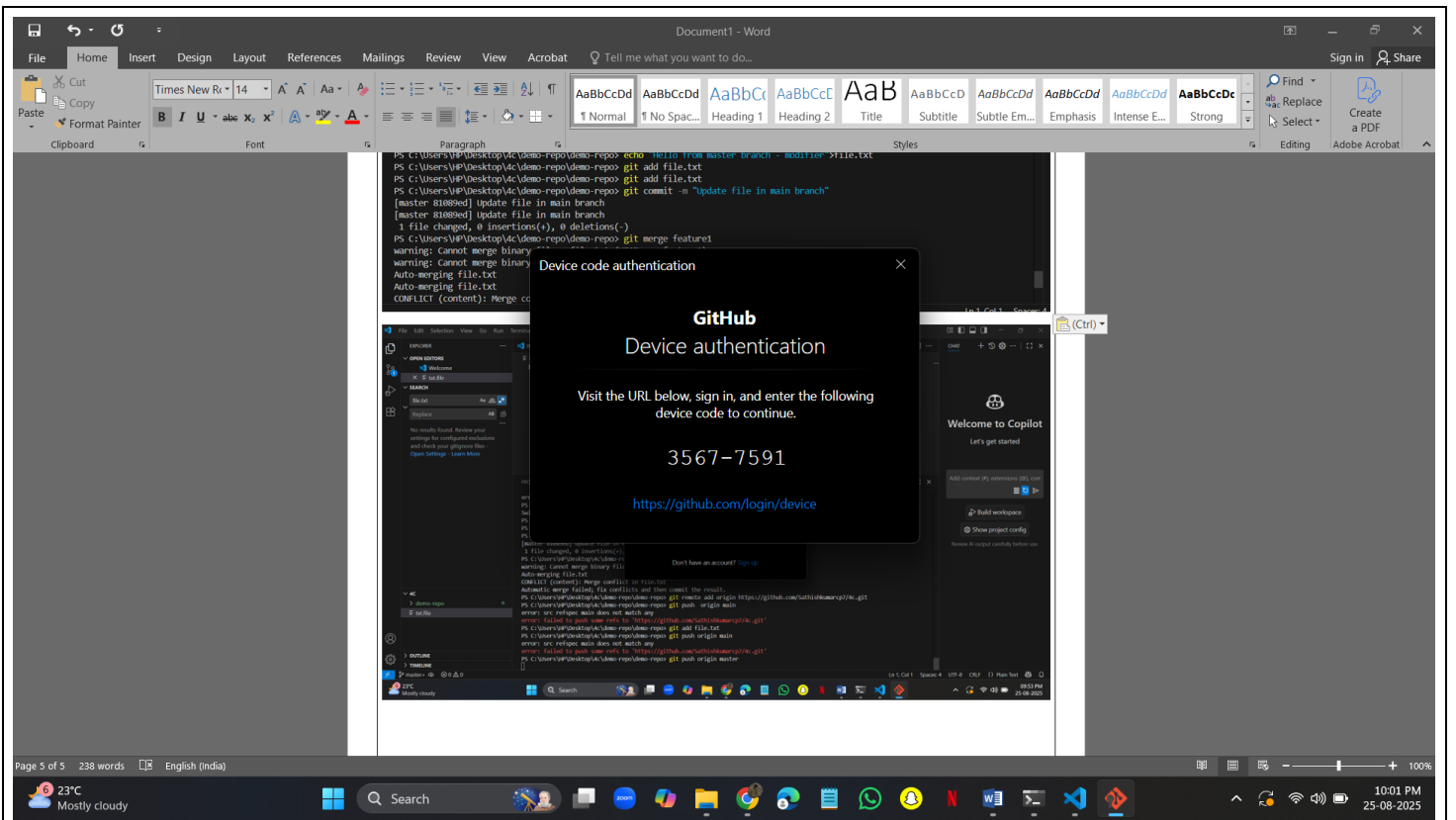
The screenshot shows the Visual Studio Code (VS Code) interface. The main editor area displays a terminal window with the following output:

```
error: pathspec 'main' did not match any file(s) known to git
PS C:\Users\HP\Desktop\4c\demo-repo> git push origin main
Switched to branch 'master'
PS C:\Users\HP\Desktop\4c\demo-repo> git push origin main
PS C:\Users\HP\Desktop\4c\demo-repo> git push origin main
PS C:\Users\HP\Desktop\4c\demo-repo> git push origin main
[master 81089ed] Update file in main
1 file changed, 0 insertions(+), 0 deletions(-)
PS C:\Users\HP\Desktop\4c\demo-repo> git push origin master
warning: Cannot merge binary files: file.txt
Auto-merging file.txt
CONFLICT (content): Merge conflict in file.txt
Automatic merge failed; fix conflicts and then commit the result.
PS C:\Users\HP\Desktop\4c\demo-repo\demo-repo> git remote add origin https://github.com/Sathishkumarc7/4c.git
PS C:\Users\HP\Desktop\4c\demo-repo\demo-repo> git push origin main
error: src refspec main does not match any
error: failed to push some refs to 'https://github.com/Sathishkumarc7/4c.git'
PS C:\Users\HP\Desktop\4c\demo-repo\demo-repo> git add file.txt
PS C:\Users\HP\Desktop\4c\demo-repo\demo-repo> git push origin main
error: src refspec main does not match any
error: failed to push some refs to 'https://github.com/Sathishkumarc7/4c.git'
PS C:\Users\HP\Desktop\4c\demo-repo\demo-repo> git push origin master
```

A "Connect to GitHub" dialog box is open in the center of the screen, prompting the user to sign in. It offers two options: "Sign in with your browser" (highlighted in blue) and "Sign in with a code". Below these options, it says "Don't have an account? Sign up".

On the right side of the interface, the "Welcome to Copilot" sidebar is visible. It includes a "Let's get started" message, a search bar for "Add context (#), extensions (@), com...", and buttons for "Build workspace" and "Show project config". A note at the bottom of the sidebar says "Review AI output carefully before use."

The bottom status bar shows the current file is "file.txt" at line 1, column 1, with a UTF-8 encoding and CRLF line endings. The system tray at the very bottom shows the date and time as "23°C Mostly cloudy" and "09:53 PM 25-08-2025".



5. Upload React Project to GitHub with Branching and Collaboration

Program:

App.js

```
import React from "react";
import Navbar from "../components/Navbar";
import ContactForm from "../components/Form";

function App() {
  return (
    <div className="App">
      <Navbar />
      <main style={{ padding: 20 }}>
        <h1>Welcome to MyApp</h1>
        <p>Practice branches, merges and pull requests.</p>
        <ContactForm />
      </main>
    </div>
  );
}

export default App;
```

Navbar.js

```
import React from "react";
import Navbar from "../components/Navbar";
import ContactForm from "../components/Form";

function App() {
  return (
    <div className="App">
      <Navbar />
      <main style={{ padding: 20 }}>
        <h1>Welcome to MyApp</h1>
```

<p>Practice branches, merges and pull requests.</p>

<ContactForm />

</main>

</div>

);

}

export default App;

Form.js

import React, { useState } from "react";

export default function ContactForm() {

const [name, setName] = useState("");

const [msg, setMsg] = useState("");

const handleSubmit = (e) => {

e.preventDefault();

alert(`Thanks \${name}! Message received: "\${msg}"`);

setName("");

setMsg("");

};

return (

<form id="contact" onSubmit={handleSubmit} style={{ maxWidth: 600 }}>

<div>

<label>Name</label>

<input value={name} onChange={(e)=>setName(e.target.value)} required />

</div>

<div style={{ marginTop: 8 }}>

<label>Message</label>

<textarea value={msg} onChange={(e)=>setMsg(e.target.value)} required />

```
    </div>

    <div style={{ marginTop: 8 }}>
      <button type="submit">Send</button>
    </div>
  </form>
);
}
```

.gitignore

node_modules/

build/

.env

Output:

```

Windows PowerShell
Bundles the app into static files for production.

npm test
Starts the test runner.

npm run eject
Removes this tool and copies build dependencies, configuration files
and scripts into the app directory. If you do this, you can't go back!

We suggest that you begin by typing:

cd sathish
npm start

Happy hacking!

C:\Users\HP>cd sathish

C:\Users\HP\sathish>npm start

> sathish@0.1.0 start
> react-scripts start

(node:5956) [DEP_WEBPACK_DEV_SERVER_ON_AFTER_SETUP_MIDDLEWARE] DeprecationWarning: 'onAfterSetupMiddleware' option is deprecated. Please use the 'setupMiddlewares' option.
(node:5956) --trace-deprecation ...
(node:5956) [DEP_WEBPACK_DEV_SERVER_ON_BEFORE_SETUP_MIDDLEWARE] DeprecationWarning: 'onBeforeSetupMiddleware' option is deprecated. Please use the 'setupMiddlewares' option.
Starting the development server...
Compiled successfully!

You can now view sathish in the browser.

Local:      http://localhost:3000
On Your Network:  http://10.26.82.135:3000

Note that the development build is not optimized.
To create a production build, use npm run build.

webpack compiled successfully

```

```
PS C:\Users\HP\sathish> git init -b main
warning: re-init: ignored --initial-branch=main
Reinitialized existing Git repository in C:/Users/HP/sathish/.git/
PS C:\Users\HP\sathish> git add .
warning: in the working copy of 'src/App.js', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'src/index.js', LF will be replaced by CRLF the next time Git touches it
PS C:\Users\HP\sathish> git commit -m "Initial commit: create-react-app scaffold and example components"
[master a211c00] Initial commit: create-react-app scaffold and example components
 5 files changed, 64 insertions(+), 32 deletions(-)
 create mode 100644 src/.gitignore
 create mode 100644 src/Form.js
 create mode 100644 src/Navbar.js
PS C:\Users\HP\sathish> git remote add origin https://github.com/Sathishkumarc7/sathish.git
>> git branch -M main
>> git push -u origin main
Enumerating objects: 30, done.
Counting objects: 100% (30/30), done.
Delta compression using up to 8 threads
Compressing objects: 100% (30/30), done.
Writing objects: 100% (30/30), 176.57 KiB | 6.79 MiB/s, done.
Total 30 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), done.
To https://github.com/Sathishkumarc7/sathish.git
 * [new branch]      main -> main
```

PROBLEMS OUTPUT DEBUG CONSOLE **TERMINAL** PORTS

 powershell


```
* [new branch]          main -> main
branch 'main' set up to track 'origin/main'.
PS C:\Users\HP\sathish> git pull origin main --allow-unrelated-histories
From https://github.com/Sathishkumarc7/sathish
* branch          main      -> FETCH_HEAD

Already up to date.
PS C:\Users\HP\sathish> git push -u origin main
branch 'main' set up to track 'origin/main'.
Everything up-to-date
PS C:\Users\HP\sathish> git branch -M main
PS C:\Users\HP\sathish> git push -u origin main
branch 'main' set up to track 'origin/main'.
Everything up-to-date
PS C:\Users\HP\sathish> git checkout -b feature/navbar
Switched to a new branch 'feature/navbar'
PS C:\Users\HP\sathish> git add src/components/Navbar.js
warning: could not open directory 'src/components/': No such file or directory
fatal: pathspec 'src/components/Navbar.js' did not match any files
PS C:\Users\HP\sathish> git add src/Navbar.js
PS C:\Users\HP\sathish> git commit -m "Feature: add responsive Navbar skeleton"
On branch feature/navbar
nothing to commit, working tree clean
PS C:\Users\HP\sathish> git push -u origin feature/navbar
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
```

PROBLEMS OUTPUT DEBUG CONSOLE **TERMINAL** PORTS

 powershell
 







```
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'feature/navbar' on GitHub by visiting:
remote:   https://github.com/Sathishkumarc7/sathish/pull/new/feature/navbar
remote:
To https://github.com/Sathishkumarc7/sathish.git
* [new branch]      feature/navbar -> feature/navbar
Branch 'feature/navbar' set up to track 'origin/feature/navbar'.
PS C:\Users\HP\sathish> git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
PS C:\Users\HP\sathish> git pull
Already up to date.
PS C:\Users\HP\sathish> git checkout -b feature/form
Switched to a new branch 'feature/form'
PS C:\Users\HP\sathish> git add src/Form.js
PS C:\Users\HP\sathish> git commit -m "Feature: add contact form"
On branch feature/form
nothing to commit, working tree clean
PS C:\Users\HP\sathish> git push -u origin feature/form
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'feature/form' on GitHub by visiting:
remote:   https://github.com/Sathishkumarc7/sathish/pull/new/feature/form
```

PROBLEMS OUTPUT DEBUG CONSOLE **TERMINAL** PORTS

 powershell


```
PS C:\Users\HP\Sathish> git commit -m "Feature: add contact form"
On branch feature/form
nothing to commit, working tree clean
PS C:\Users\HP\Sathish> git push -u origin feature/form
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'feature/form' on GitHub by visiting:
remote:   https://github.com/Sathishkumarc7/sathish/pull/new/feature/form
remote:
To https://github.com/Sathishkumarc7/sathish.git
 * [new branch]      feature/form -> feature/form
branch 'feature/form' set up to track 'origin/feature/form'.
PS C:\Users\HP\Sathish> git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
PS C:\Users\HP\Sathish> git pull
Already up to date.
PS C:\Users\HP\Sathish> git merge feature/navbar
Already up to date.
PS C:\Users\HP\Sathish> git push origin main
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
```

```
nothing to commit, working tree clean
PS C:\Users\HP\sathish> git branch
  feature/form
  feature/navbar
* main
PS C:\Users\HP\sathish> git branch -a
  feature/form
  feature/navbar
* main
remotes/origin/HEAD -> origin/main
remotes/origin/feature/form
remotes/origin/feature/navbar
remotes/origin/main
PS C:\Users\HP\sathish> git log --online --graph --all
fatal: unrecognized argument: --online
PS C:\Users\HP\sathish> git remote -v
origin https://github.com/Sathishkumarc7/sathish.git (fetch)
origin https://github.com/Sathishkumarc7/sathish.git (push)
PS C:\Users\HP\sathish> git remote remove origin
PS C:\Users\HP\sathish> git remote add origin https://github.com/Sathishkumarc7/sathish.git
```


Sathishkumarc7 / sathish

Type **7** to search

<> Code

Issues

Pull requests

Actions

Projects

Wiki

Security

Insights

Settings

Compare changes

Compare changes across branches, commits, tags, and more below. If you need to, you can also [compare across forks](#).

base: main

←

compare: main

Choose different branches or forks above to discuss and review changes. [Learn about pull requests](#)

Create pull request

Compare and review just about anything

Branches, tags, commit ranges, and time ranges. In the same repository and across forks.

| Example comparisons | |
|--------------------------------|--------------|
| <div></div> feature/navbar | 1 hour ago |
| <div></div> feature/form | 1 hour ago |
| <div></div> main@(1day)...main | 24 hours ago |

6(a). Node.js using the Express framework

Program:

app.js

```
// Import express
```

```
const express = require("express");
```

```
// Create an express application
```

```
const app = express();
```

```
// Define a route for '/'
```

```
app.get("/", (req, res) => {
```

```
    res.send("Hello World");
```

```
});
```

```
// Start the server on port 3000
```

```
app.listen(3000, () => {
```

```
    console.log("Server is running on http://localhost:3000");
```

```
});
```

Output:

```
Command Prompt
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C:\Users\HP>node -v
v22.18.0

C:\Users\HP>npm -v
10.9.3

C:\Users\HP>mkdir express-6a

C:\Users\HP>cd express-6a

C:\Users\HP\express-6a>npm init -y
Wrote to C:\Users\HP\express-6a\package.json:

{
  "name": "express-6a",
  "version": "1.0.0",
  "description": "",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC"
}

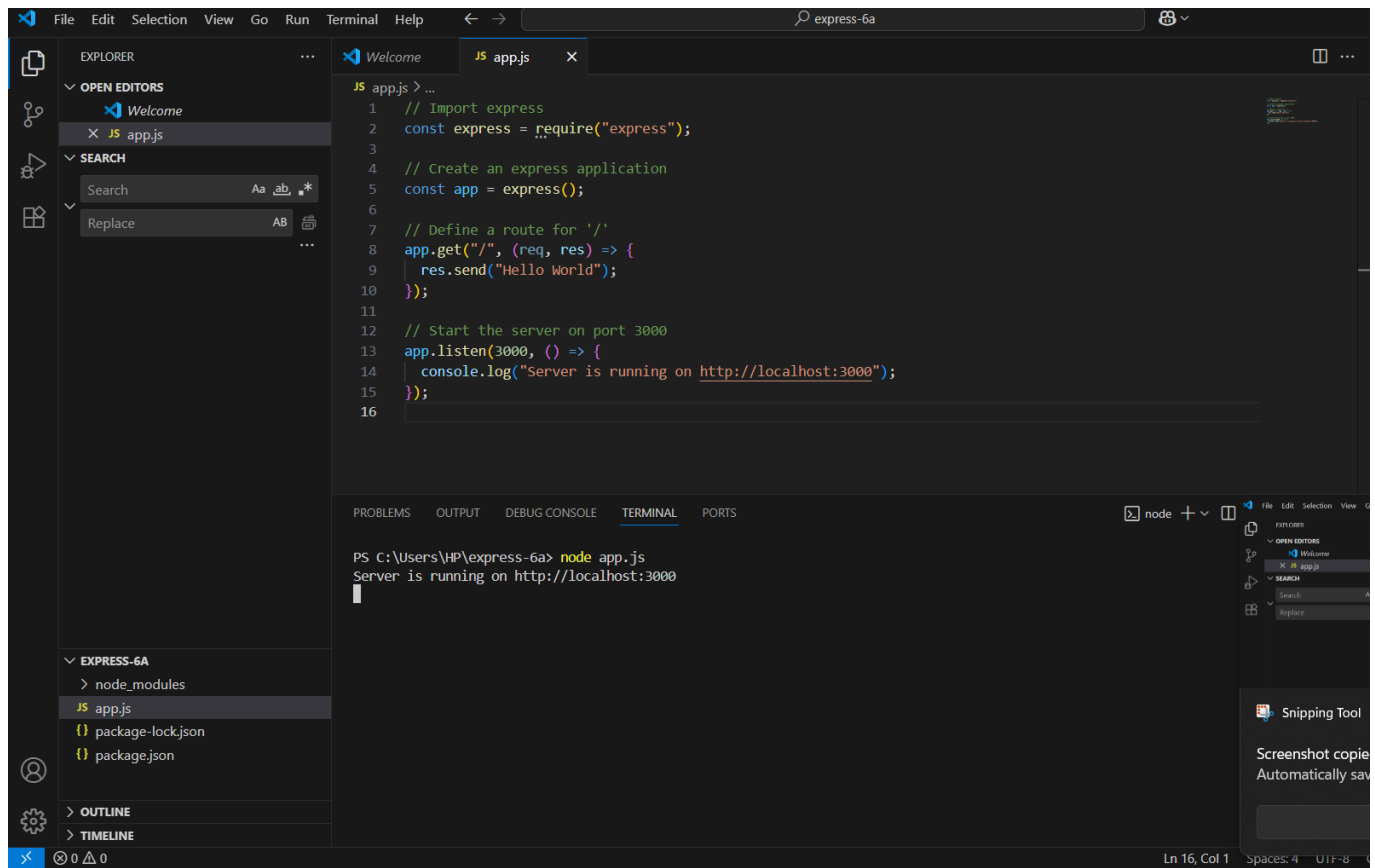
C:\Users\HP\express-6a>npm install express

added 67 packages, and audited 68 packages in 5s

14 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities

C:\Users\HP\express-6a>
```





Hello World

6(b). Multiple routes using Express.js

PROGRAM:

app.js

```
// Import express
```

```
const express = require("express");
```

```
// Create app
```

```
const app = express();
```

```
// Define routes
```

```
app.get("/", (req, res) => {
```

```
    res.send("<h1>Welcome to Home Page</h1><p>This is the main route.</p>");
```

```
});
```

```
app.get("/about", (req, res) => {
```

```
    res.send("<h1>About Us</h1><p>This website is built using Express.js.</p>");
```

```
});
```

```
app.get("/contact", (req, res) => {
```

```
    res.send("<h1>Contact Us</h1><p>Email: support@example.com</p>");
```

```
});
```

```
app.get("/services", (req, res) => {
```

```
    res.send("<h1>Our Services</h1><ul><li>Web Development</li><li>App Development</li></ul>");
```

```
});
```

```
// Start the server
```

```
const PORT = 3000;

app.listen(PORT, () => {

  console.log(`Server running at http://localhost:${PORT}`);

});
```

Output:

```
Command Prompt
(c) Microsoft Corporation. All rights reserved.

C:\Users\HP>node -v
v22.18.0

C:\Users\HP>npm -v
10.9.3

C:\Users\HP>mkdir express-multi-routes

C:\Users\HP>cd express-multi-routes

C:\Users\HP\express-multi-routes>npm init -y
Wrote to C:\Users\HP\express-multi-routes\package.json:

{
  "name": "express-multi-routes",
  "version": "1.0.0",
  "description": "",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC"
}

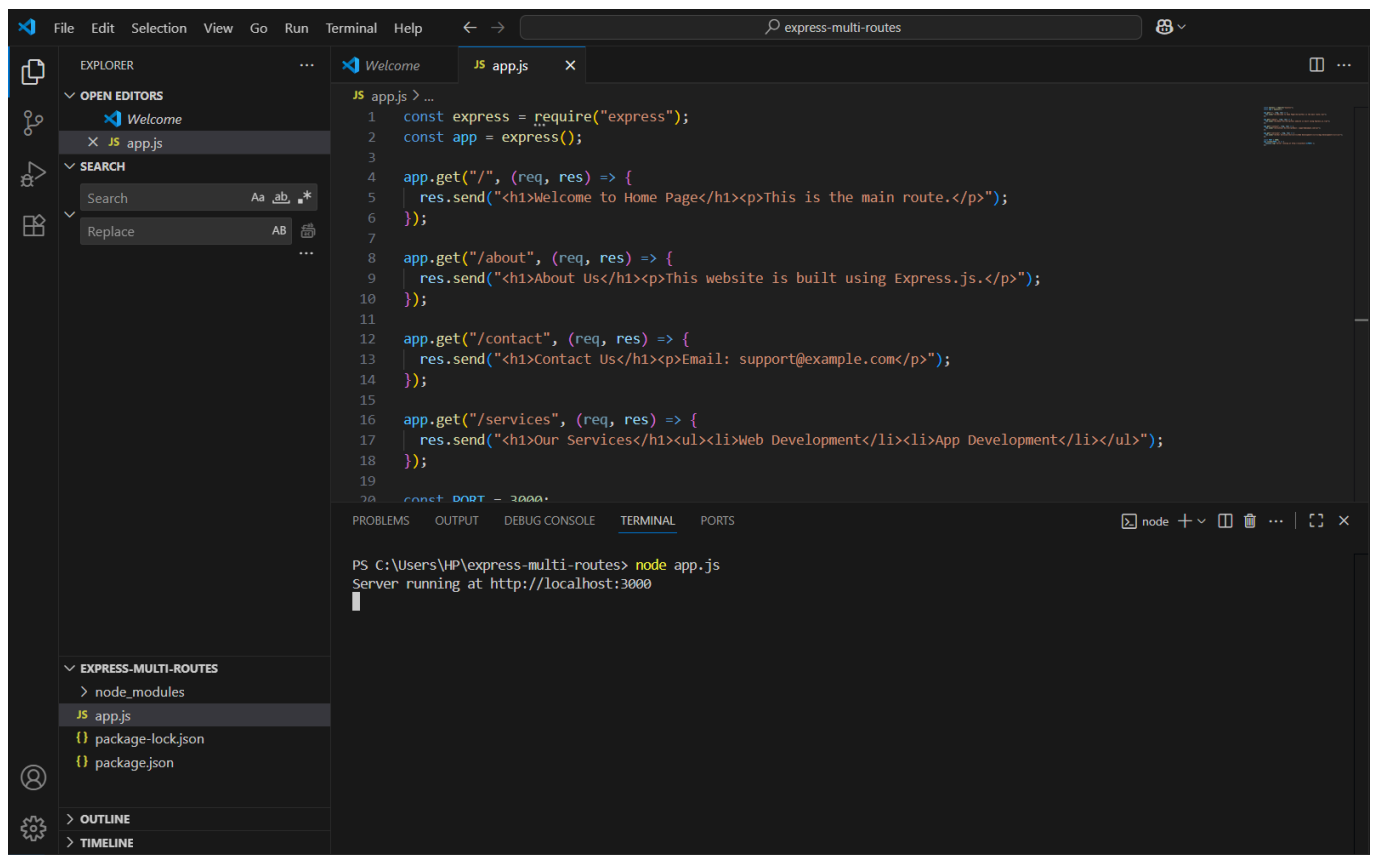
C:\Users\HP\express-multi-routes>npm install express

added 67 packages, and audited 68 packages in 4s

14 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities

C:\Users\HP\express-multi-routes>
```



The screenshot shows the Visual Studio Code editor with the 'express-multi-routes' project open. The Explorer sidebar on the left shows the file structure with 'node_modules', 'package-lock.json', and 'package.json'. The main editor area displays the 'app.js' file, which contains the following code:

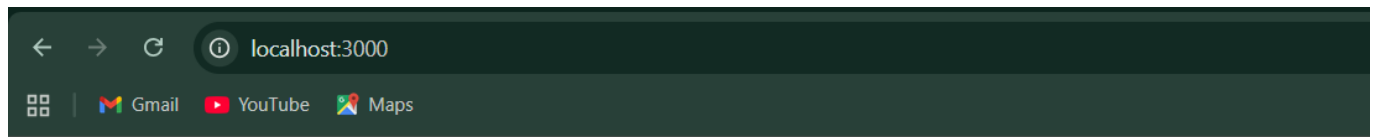
```
1 const express = require("express");
2 const app = express();
3
4 app.get("/", (req, res) => {
5   res.send("<h1>Welcome to Home Page</h1><p>This is the main route.</p>");
6 });
7
8 app.get("/about", (req, res) => {
9   res.send("<h1>About Us</h1><p>This website is built using Express.js.</p>");
10 });
11
12 app.get("/contact", (req, res) => {
13   res.send("<h1>Contact Us</h1><p>Email: support@example.com</p>");
14 });
15
16 app.get("/services", (req, res) => {
17   res.send("<h1>Our Services</h1><ul><li>Web Development</li><li>App Development</li></ul>");
18 });
19
20 const PORT = 3000;
```

The bottom panel shows the 'TERMINAL' tab with the following output:

```
PS C:\Users\HP\express-multi-routes> node app.js
Server running at http://localhost:3000
```

When you visit

- <http://localhost:3000/> → **Welcome to Home Page**
- <http://localhost:3000/about> → **About Us page**
- <http://localhost:3000/contact> → **Contact page**
- <http://localhost:3000/services> → **Our Services page**



Welcome to Home Page

This is the main route.



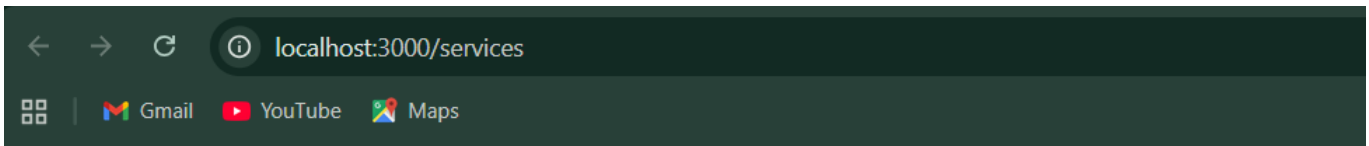
About Us

This website is built using Express.js.



Contact Us

Email: support@example.com



Our Services

- Web Development
- App Development

6(c). Express.js that runs a server and prints "**Hello World**" in the browser console

PROGRAM:

Server.js

```
// Import Express
const express = require("express");
const app = express();

// Handle favicon request (avoids 404 error)
app.get("/favicon.ico", (req, res) => res.status(204));

// Define a route
app.get("/", (req, res) => {
  res.send(`
    <html>
      <head>
        <title>Hello World App</title>
      </head>
      <body>
        <h1>Check your browser console!</h1>
        <script>
          console.log("Hello World");
        </script>
      </body>
    </html>
  `);
});

// Start the server
app.listen(3000, () => {
  console.log("Server running at http://localhost:3000");
});
```

OUTPUT:

```
Command Prompt
Microsoft Windows [Version 10.0.22631.5768]
(c) Microsoft Corporation. All rights reserved.

C:\Users\HP>node -v
v22.18.0

C:\Users\HP>mkdir express-hello

C:\Users\HP>cd express-hello

C:\Users\HP\express-hello>npm init -y
Wrote to C:\Users\HP\express-hello\package.json:

{
  "name": "express-hello",
  "version": "1.0.0",
  "description": "",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC"
}

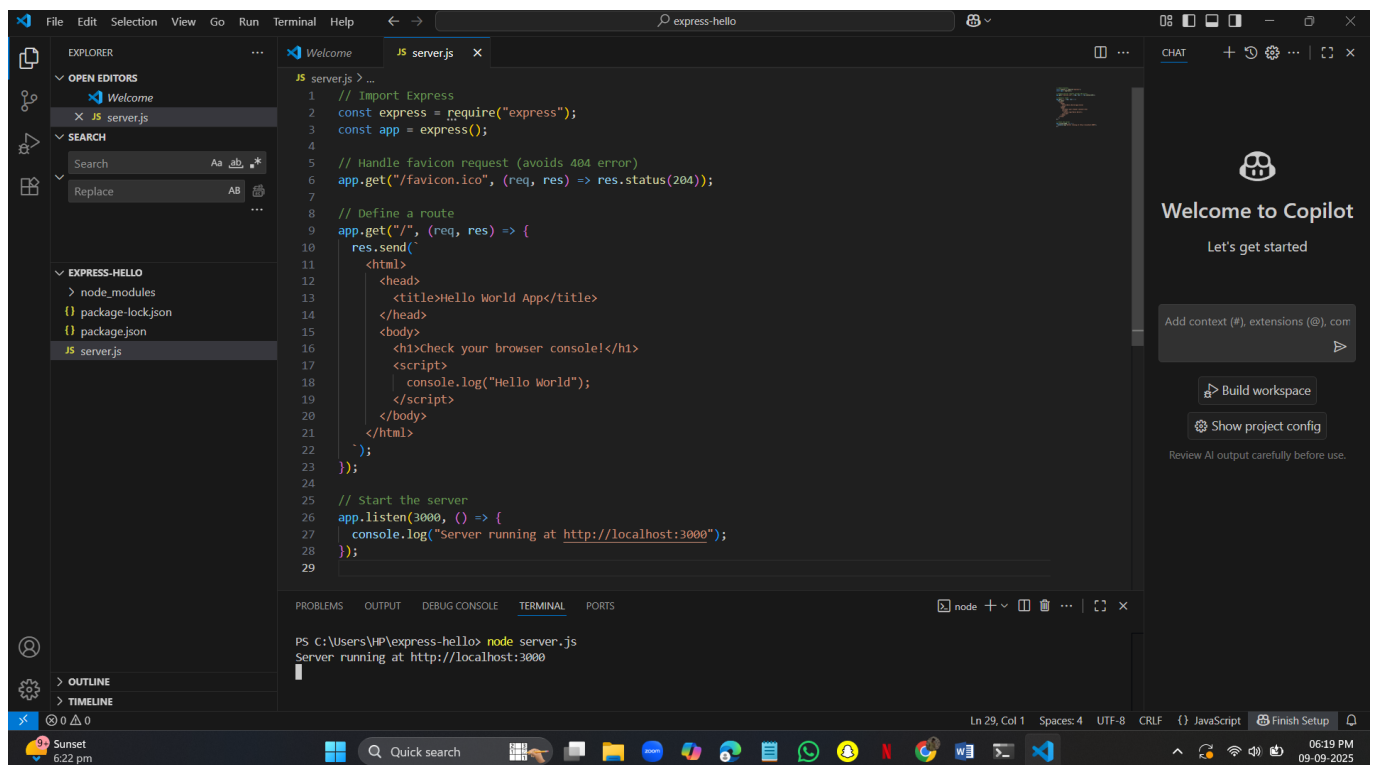
C:\Users\HP\express-hello>npm install express

added 68 packages, and audited 69 packages in 4s

16 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities

C:\Users\HP\express-hello>
```

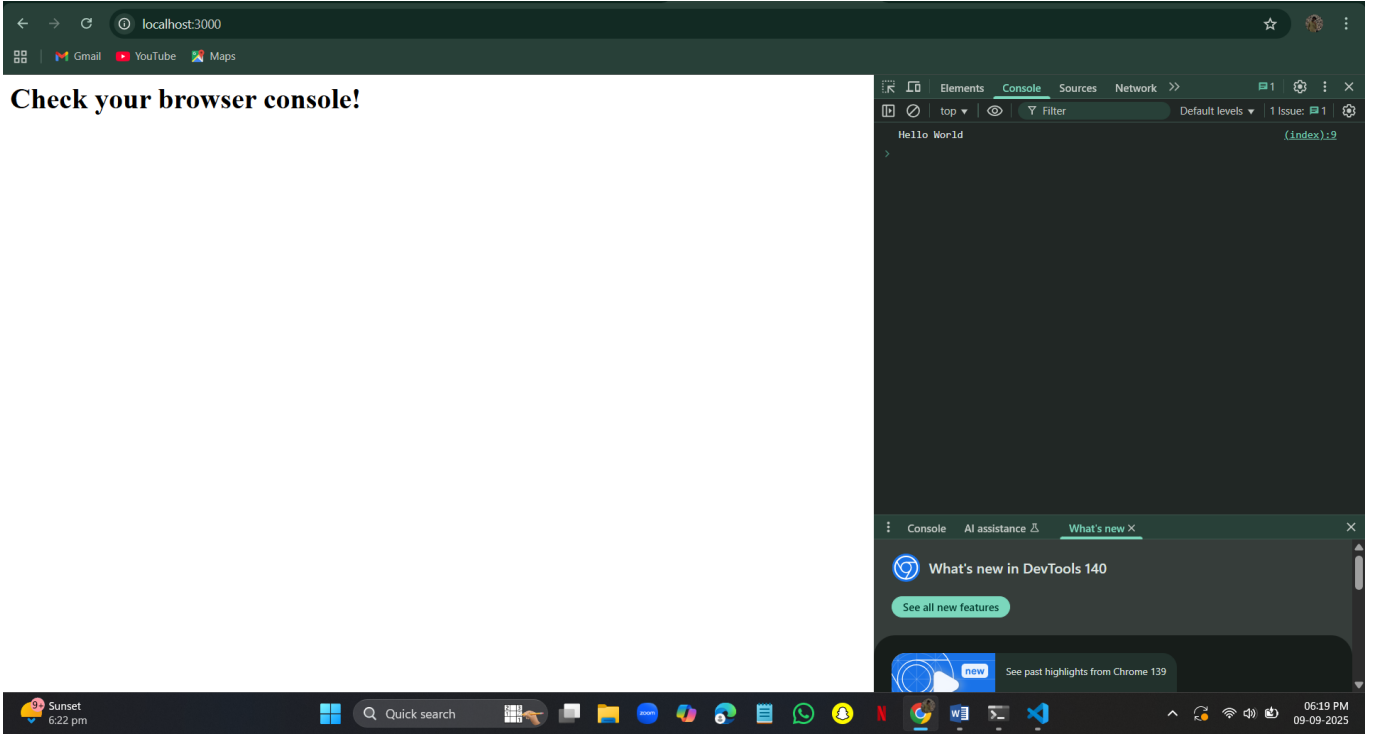


```
File Edit Selection View Go Run Terminal Help
express-hello

EXPLORER
  OPEN EDITORS
    Welcome
    JS server.js
  SEARCH
    Search
    Replace
  EXPRESS-HELLO
    > node_modules
    () package-lock.json
    () package.json
    JS server.js

JS server.js
1 // Import Express
2 const express = require("express");
3 const app = express();
4
5 // Handle favicon request (avoids 404 error)
6 app.get("/favicon.ico", (req, res) => res.status(204));
7
8 // Define a route
9 app.get("/", (req, res) => {
10   res.send(`
11     <html>
12       <head>
13         <title>Hello World App</title>
14       </head>
15       <body>
16         <h1>Check your browser console!</h1>
17         <script>
18           console.log("Hello World");
19         </script>
20       </body>
21     </html>
22   `);
23 });
24
25 // Start the server
26 app.listen(3000, () => {
27   console.log("Server running at http://localhost:3000");
28 });
29

TERMINAL
PS C:\Users\HP\express-hello> node server.js
Server running at http://localhost:3000
```



6(d). Express.js that implements CRUD operations (Create, Read, Update, Delete) for managing data

PROGRAM:

CurdApp.js

```
// Import express
```

```
const express = require("express");
```

```
const app = express();
```

```
// Middleware to parse JSON
```

```
app.use(express.json());
```

```
// Temporary in-memory "database"
```

```
let items = [
```

```
  { id: 1, name: "Item One" },
```

```
  { id: 2, name: "Item Two" }
```

```
];
```

```
// 1. READ - Get all items
```

```
app.get("/items", (req, res) => {
```

```
  res.json(items);
```

```
});
```

```
// 2. CREATE - Add a new item
```

```
app.post("/items", (req, res) => {
```

```
  const newItem = {
```

```
    id: items.length + 1,
    name: req.body.name
  };
  items.push(newItem);
  res.status(201).json(newItem);
});

// 3. UPDATE - Update an item by ID
app.put("/items/:id", (req, res) => {
  const id = parseInt(req.params.id);
  const item = items.find(i => i.id === id);

  if (!item) {
    return res.status(404).json({ message: "Item not found" });
  }

  item.name = req.body.name;
  res.json(item);
});

// 4. DELETE - Remove an item by ID
app.delete("/items/:id", (req, res) => {
  const id = parseInt(req.params.id);
  items = items.filter(i => i.id !== id);
  res.json({ message: "Item deleted successfully" });
});
```

```
});
```

```
// Start server
```

```
app.listen(3000, () => {
```

```
  console.log("Server is running on http://localhost:3000");
```

```
});
```

OUTPUT:

```
Command Prompt
(c) Microsoft Corporation. All rights reserved.

C:\Users\HP>node -v
v22.18.0

C:\Users\HP>npm -v
10.9.3

C:\Users\HP>mkdir express-crud

C:\Users\HP>cd express-crud

C:\Users\HP\express-crud>npm init -y
Wrote to C:\Users\HP\express-crud\package.json:

{
  "name": "express-crud",
  "version": "1.0.0",
  "description": "",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC"
}

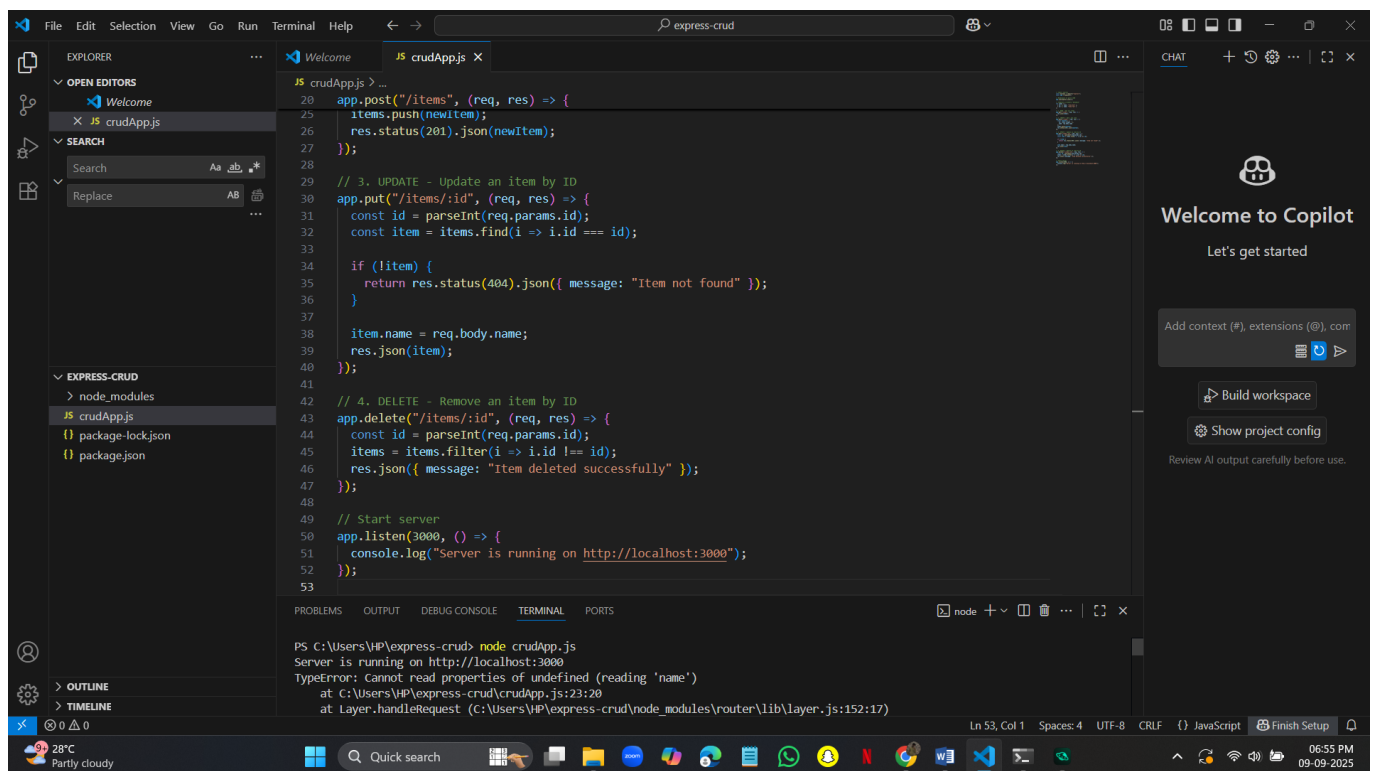
C:\Users\HP\express-crud>npm install express

added 68 packages, and audited 69 packages in 4s

16 packages are looking for funding
  run 'npm fund' for details

found 0 vulnerabilities

C:\Users\HP\express-crud>
```



The screenshot shows the Visual Studio Code interface with the following components:

- EXPLORER:** Shows the project structure with files `crudApp.js`, `package-lock.json`, and `package.json`.
- EDITOR:** Displays the `crudApp.js` file with the following JavaScript code:

```
20 app.post("/items", (req, res) => {
21   items.push(newItem);
22   res.status(201).json(newItem);
23 });
24
25 // 3. UPDATE - Update an item by ID
26 app.put("/items/:id", (req, res) => {
27   const id = parseInt(req.params.id);
28   const item = items.find(i => i.id === id);
29
30   if (!item) {
31     return res.status(404).json({ message: "Item not found" });
32   }
33
34   item.name = req.body.name;
35   res.json(item);
36 });
37
38 // 4. DELETE - Remove an item by ID
39 app.delete("/items/:id", (req, res) => {
40   const id = parseInt(req.params.id);
41   items = items.filter(i => i.id !== id);
42   res.json({ message: "Item deleted successfully" });
43 });
44
45 // Start server
46 app.listen(3000, () => {
47   console.log("Server is running on http://localhost:3000");
48 });
```
- TERMINAL:** Shows the command `node crudApp.js` and the following output:

```
PS C:\Users\HP\express-crud> node crudApp.js
Server is running on http://localhost:3000
TypeError: Cannot read properties of undefined (reading 'name')
    at C:\Users\HP\express-crud\crudApp.js:23:20
    at Layer.handleRequest (C:\Users\HP\express-crud\node_modules\router\lib\layer.js:152:17)
```
- CHAT:** Displays the "Welcome to Copilot" message.

If you have run means install POSTMAN API

